ESAT PRACTICE TEST SERIES

PHYSICS + BIOLOGY

LEVEL 1 · STANDARD

Physics + Biology

Mathematics 1 + Physics + Biology

Representative ESAT-style breadth with routine and multi-step applications.

MODULE

Mathematics 1

MODULE

Physics

MODULE

Biology

FULL SOLUTION BOOK

ThrivingScholars

81 preserved questions · three verified module keys

COMBINED ANSWER KEYS

Physics + Biology · Level 1 Standard

This book compiles three existing 27-question modules into one 81-question pathway. No new questions have been generated and no source solution has been rewritten.

Composition: Mathematics 1 + Physics + Biology. Each module retains its original question order, answer and worked solution.

Mathematics 1

Engineering Mock Test 3 · preserved Mathematics 1 module

Q1 C	Q2 B	Q3 E	Q4 C	Q5 D
Q6 D	Q7 E	Q8 A	Q9 A	Q10 D
Q11 D	Q12 C	Q13 B	Q14 A	Q15 B
Q16 E	Q17 E	Q18 E	Q19 D	Q20 E
Q21 D	Q22 D	Q23 C	Q24 D	Q25 D
Q26 A	Q27 C			

Physics

Engineering Mock Test 3 · preserved Physics module

Q1 D	Q2 E	Q3 B	Q4 A	Q5 E
Q6 D	Q7 F	Q8 E	Q9 A	Q10 C
Q11 D	Q12 F	Q13 E	Q14 H	Q15 C
Q16 F	Q17 A	Q18 H	Q19 D	Q20 A
Q21 D	Q22 F	Q23 E	Q24 B	Q25 H
Q26 G	Q27 C			

Biology

Preserved Biology Standard source set

Q1 D	Q2 C	Q3 C	Q4 G	Q5 C
Q6 F	Q7 E	Q8 F	Q9 C	Q10 D
Q11 B	Q12 E	Q13 B	Q14 D	Q15 C
Q16 E	Q17 E	Q18 H	Q19 B	Q20 C
Q21 E	Q22 D	Q23 B	Q24 B	Q25 C
Q26 F	Q27 E			

LEVEL 1 · STANDARD

Mathematics 1

27 QUESTIONS · COMPLETE SOLUTIONS

Engineering Mock Test 3 · preserved Mathematics 1 module. Question wording, option order, answer and solution method are unchanged.

M1-01

Mock Test 3 - Mathematics 1

Answer C

QUESTION 1

There are 8 consecutive integers. If the sum of the first three is 63, what is the largest one?

A 17

B 25

C 27

D 30

E 31

WORKED SOLUTION

Let the first integer be n. Then $n + (n + 1) + (n + 2) = 63$, so $3n + 3 = 63$, giving $n = 20$. The largest of the eight integers is 27.

M1-02

Mock Test 3 - Mathematics 1 | Answer **B**

QUESTION 2

Joshua is 10 years old and his father is 42 years old. After how many years, his father's age is three times Joshua's age?

- **A** 5
- **B** 6
- **C** 7
- **D** 8
- **E** 9

WORKED SOLUTION

After t years, $42 + t = 3(10 + t)$. Hence $42 + t = 30 + 3t$, so $t = 6$.

M1-03

Mock Test 3 - Mathematics 1 | Answer **E**

QUESTION 3

N is a 7-digit number consists of 2 or 3. There are more 2s than 3s in N. If N is divisible by both 3 and 4, what is the remainder of N when it is divided by 7?

- **A** 1
- **B** 2
- **C** 3

D 4

E 5

WORKED SOLUTION

Let k be the number of 3s. The digit sum is $14 + k$, so divisibility by 3 gives $k = 1$. Divisibility by 4 forces the last two digits to be 32. Thus $N = 2222232$, and $N \equiv 5 \pmod{7}$.

M1-04

Mock Test 3 - Mathematics 1

Answer **C**

QUESTION 4

The preliminary round of a violin competition is divided into four rounds. According to the rule, the average score of a percipient of these four rounds must be not less than 96 points in order to enter the finals. Shaelyn's scores in the first three rounds are 95, 97, and 94. What is the minimum score that she needs to get on the fourth round in order to enter the finals? (Hope Cup Math Competition, China)

A 96

B 97

C 98

D 99

E 100

WORKED SOLUTION

The total score needed is $4 \times 96 = 384$. The first three scores total $95 + 97 + 94 = 286$, so the fourth score must be at least $384 - 286 = 98$.

M1-05

Mock Test 3 - Mathematics 1

Answer **D**

QUESTION 5

A haunted house has 8 windows. In how many ways can George the Ghost enter the house by one window and leave by a different window?

- **A** 28
- **B** 32
- **C** 40
- **D** 56
- **E** 63

WORKED SOLUTION

There are 8 choices for the entry window and then 7 different choices for the exit window, giving $8 \times 7 = 56$.

M1-06

Mock Test 3 - Mathematics 1

Answer **D**

QUESTION 6

Jackie bought a bottle (30 oz) of 90% alcohol from pharmacy to make homemade disinfectant spray. How many ounces of water should he add in order to dilute it into 75% alcohol? (Hint: The amount of alcohol remains the same.)

A 3

B 4

C 5

D 6

E 7

WORKED SOLUTION

The amount of alcohol is $30 \times 0.90 = 27$ oz. If the final concentration is 75%, then $0.75\mathrm{T} = 27$, so $\mathrm{T} = 36$. Water added: $36 - 30 = 6$ oz.

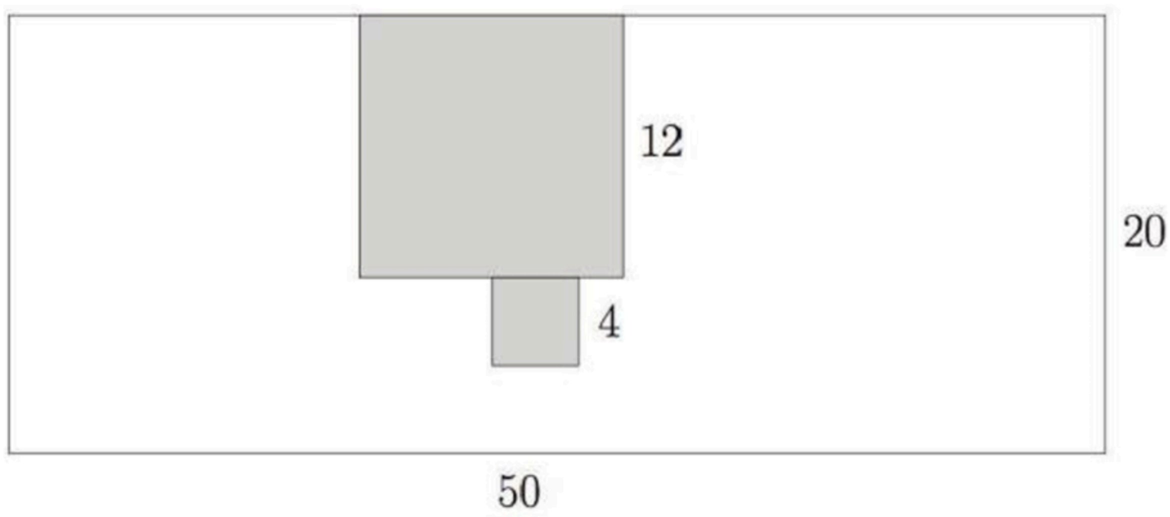
M1-07

Mock Test 3 - Mathematics 1

Answer E

QUESTION 7

Cut a 12×12 square and a 4×4 square from a 50×20 rectangle paper. What is the perimeter of the remaining part?



A 128

B 140

C 152

D 160

E 172

WORKED SOLUTION

The original rectangle perimeter is $2(50+20)=140$. The 12×12 notch increases the perimeter by 24, and the attached 4×4 cut-out adds another 8. Total perimeter $= 140+32=172$.

M1-08

Mock Test 3 - Mathematics 1

Answer A

QUESTION 8

Jasmine keeps adding numbers one by one from the following list

1, 3, 5, 7, ⋯ .

as $1+3=4$, $1+3+5=9$, $1+3+5+7=16$, etc. At one point, she got 379 and realized that one number is missed in the addition. What is the sum of digits of the missing number?

A 3

B 6

C 8

D 9

E 11

WORKED SOLUTION

The sum of the first n odd numbers is n^2. Since 379 is just below $20^2 = 400$, the missing number is $400 - 379 = 21$. The digit sum is $2 + 1 = 3$.

M1-09

Mock Test 3 - Mathematics 1

Answer A

QUESTION 9

$\overline{ab}$ and $\overline{cd}$ are two 2-digit numbers. If

$\overline{ab} - \overline{cd} = 24$

and

$\overline{1ab} - \overline{cd1} = 15,$

what is $a + d$?

A 5

B 8

C 10

D 11

E 12

WORKED SOLUTION

Using place value, $\overline{ab} = 10a + b$ and $\overline{cd} = 10c + d$. Solving the two given equations gives $a + d = 5$.

M1-10

Mock Test 3 - Mathematics 1

Answer **D**

QUESTION 10

The speed of a train is 15 meters per second. If it takes the train 10 seconds to pass a lamp post, how long does it take the train to completely pass a 300-meter long tunnel, in seconds? (Hint: first figure out the length of the train based on the information related to the lamppost)

- **A** 10
- **B** 20
- **C** 25
- **D** 30
- **E** 35

WORKED SOLUTION

The train length is $15 \times 10 = 150$ m. To pass the tunnel completely, it travels $300 + 150 = 450$ m. Time $= 450/15 = 30$ s.

M1-11

Mock Test 3 - Mathematics 1

Answer **D**

QUESTION 11

How many pairs of positive integers (x, y) satisfy that $\frac{x}{3} = \frac{4}{y}$?

A 3

B 4

C 5

D 6

E 7

WORKED SOLUTION

From $x/3 = 4/y$, we get $xy = 12$. The positive factor pairs are $(1, 12), (2, 6), (3, 4), (4, 3), (6, 2), (12, 1)$, so there are 6 pairs.

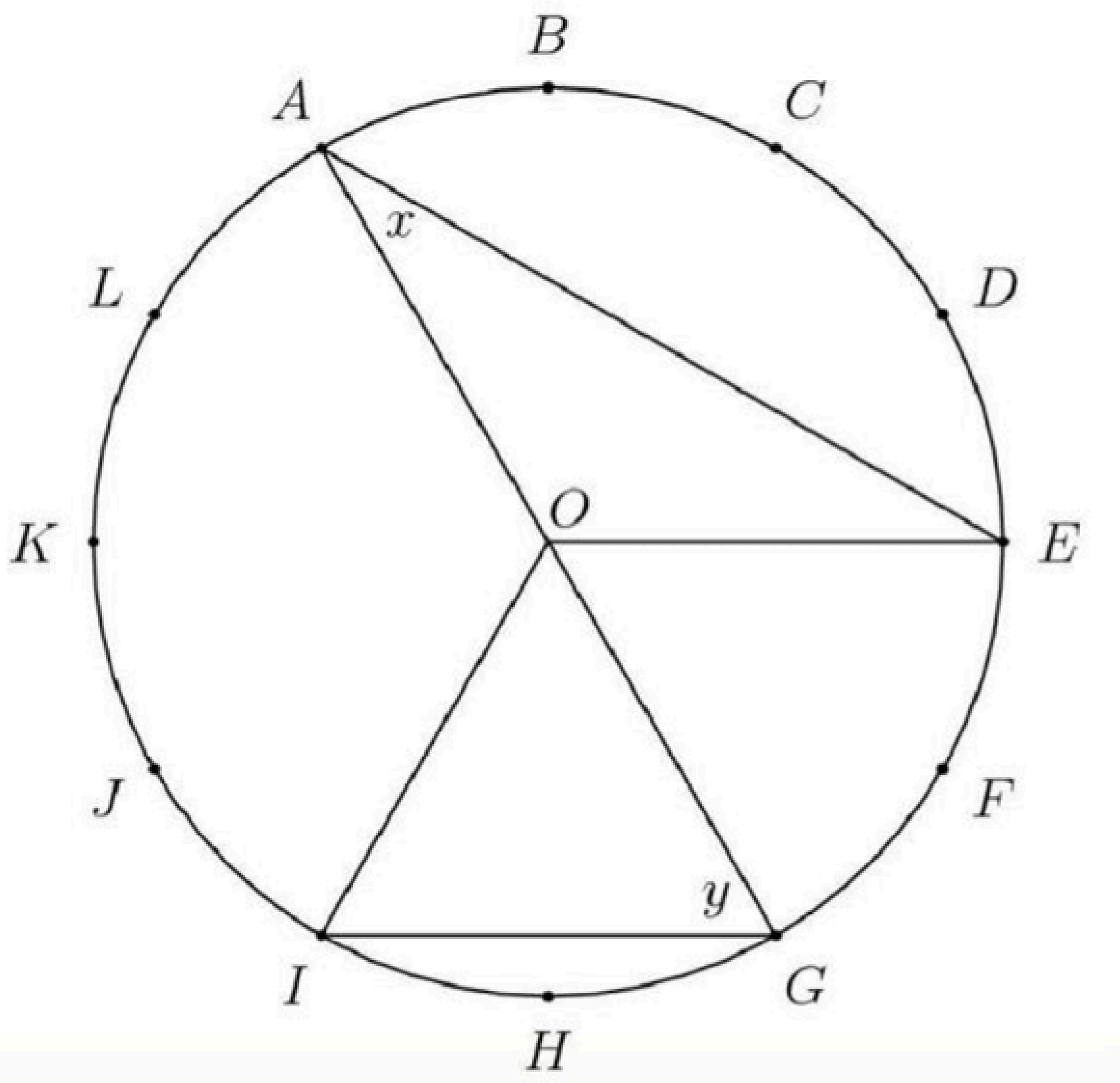
M1-12

Mock Test 3 - Mathematics 1

Answer C

QUESTION 12

The circumference of the circle with center O is divided into 12 equal arcs, marked the letters A through L as seen below. What is the number of degrees in the sum of the angles x and y?



A 75

B 80

C 90

D 120

E 150

WORKED SOLUTION

The circle is divided into 12 equal arcs, so each arc is 30°. Using central and inscribed angle relationships from the diagram gives $x + y = 90^\circ$.

M1-13

Mock Test 3 - Mathematics 1

Answer B

QUESTION 13

If for whole numbers x, y and z,

$360 = 2^x \times 3^y \times 5^z$,

what is $x + y + z$? Here $a^n = a \times a \times \cdots \times a$.

For example, $100 = 2 \times 2 \times 5 \times 5 = 2^2 \times 5^2$.

A 5

B 6

C 7

D 8

E 9

WORKED SOLUTION

Prime factorise: $360 = 36 \times 10 = 2^2 \times 3^2 \times 2 \times 5 = 2^3 \times 3^2 \times 5^1$. Hence $x + y + z = 3 + 2 + 1 = 6$.

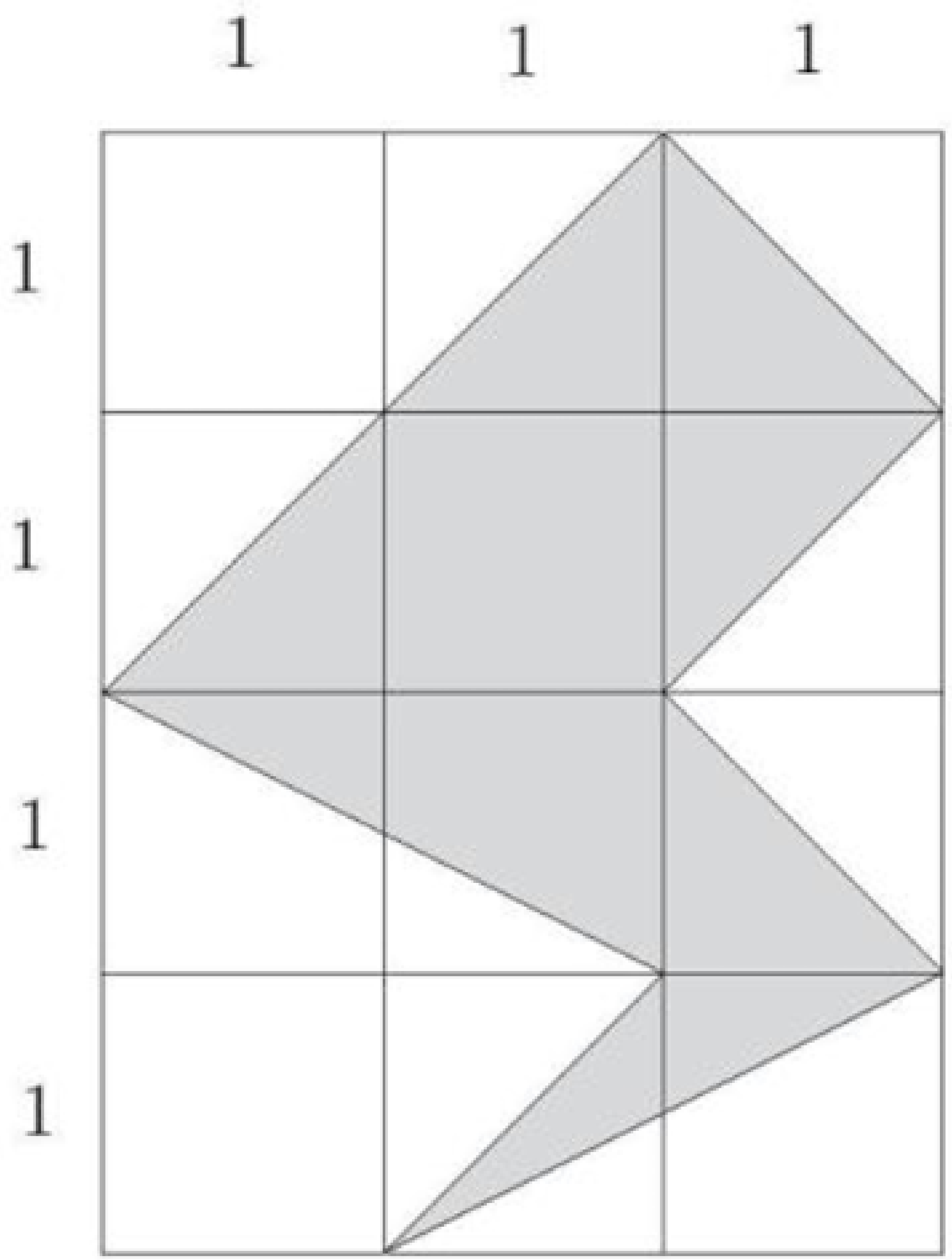
M1-14

Mock Test 3 - Mathematics 1

Answer **A**

QUESTION 14

What is the area of the shaded region in the following 3×4 grid?



A 5

B 6

C 7

D 8

E 9

WORKED SOLUTION

Counting the shaded polygon by subtracting the surrounding right triangles from the 3×4 rectangle gives shaded area 5.

M1-15

Mock Test 3 - Mathematics 1

Answer B

QUESTION 15

Randomly choose 3 different numbers from the set $\{1, 3, 5, 7, 9\}$. What is the probability that the three selected numbers could be side lengths of a triangle?

A $\frac{1}{5}$

B $\frac{3}{10}$

C $\frac{1}{3}$

D $\frac{2}{5}$

E $\frac{1}{2}$

WORKED SOLUTION

There are $\binom{5}{3} = 10$ triples. The valid triangle triples are $(3, 5, 7), (3, 7, 9), (5, 7, 9)$, so the probability is $3/10$.

M1-16

Mock Test 3 - Mathematics 1

Answer E

QUESTION 16

Marley practices exactly one sport each day of the week. She runs three days a week but never on two consecutive days. On Monday she plays basketball and two days later golf. She swims and plays tennis, but she never plays tennis the day after running or swimming. Which day of the week does Marley swim?

- **A** Sunday
- **B** Tuesday
- **C** Thursday
- **D** Friday
- **E** Saturday

WORKED SOLUTION

Monday is basketball and Wednesday is golf. Applying the non-consecutive running rule and the condition that tennis is not immediately after running or swimming leaves Saturday as the swimming day.

M1-17

Mock Test 3 - Mathematics 1

Answer E

QUESTION 17

Three toy cars run around a round track. It takes them 8, 10 and 12 seconds to finish one full circle respectively. If three cars start together from the same point and run toward the same direction, after how many seconds will they get back to the starting point together for the first time?

A 20

B 40

C 60

D 80

E 120

WORKED SOLUTION

The cars return together after the least common multiple of $8, 10, 12$, which is 120 seconds.

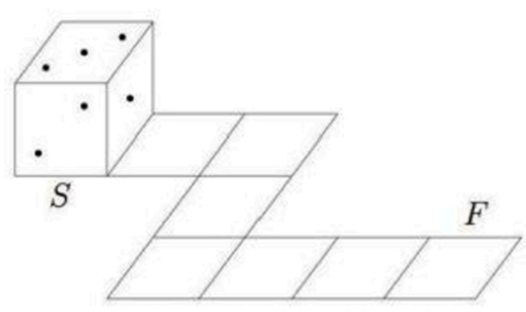
M1-18

Mock Test 3 - Mathematics 1

Answer E

QUESTION 18

Opposite faces of a die always add to 7. A die rolls on a circuit as represented below. At the starting point (S) the top face is 3. Which will be the top face at the final point (F)?



A 2

B 3

C 4

D 5

E 6

F 7

G 8

H 9

WORKED SOLUTION

Track the die through the path, using opposite faces summing to 7 after each roll. The final top face is 6.

M1-19

Mock Test 3 - Mathematics 1

Answer D

QUESTION 19

How many numbers x satisfy that $(x^2 - 2)^2 = 4$?

A 0

B 1

C 2

D 3

E 4

F 5

G 6

H 7

WORKED SOLUTION

$(x^2 - 2)^2 = 4$ gives $x^2 - 2 = \pm 2$. Thus $x^2 = 4$ or $x^2 = 0$, giving $x = -2, 0, 2$. There are 3 real numbers.

M1-20

Mock Test 3 - Mathematics 1

Answer E

QUESTION 20

For how many two-digit numbers, if you switch the tens and ones digits (e.g., 18 → 81), the new 2-digit number is 18 more than the original one?

A 3

B 4

C 5

D 6

E 7

WORKED SOLUTION

Let the number be $10a + b$. Reversing digits gives $10b + a$. The condition gives $10b + a = 10a + b + 18$, so $b - a = 2$. There are 7 possible tens digits.

M1-21 Mock Test 3 - Mathematics 1 Answer D

QUESTION 21

[AMC 8 Problem] Malcolm wants to visit Isabella after school today and knows the street where she lives but doesn't know her house number. She tells him, "My house number has two digits, and exactly three of the following four statements about it are true."

- It is prime.
- It is even.
- It is divisible by 7.
- One of its digits is 9.

This information allows Malcolm to determine Isabella's house number. What is its units digit?

A 4

B 6

C 7

D 8

E 9

WORKED SOLUTION

The number must have exactly three of the four properties. The number 98 is even, divisible by 7, has a digit 9, and is not prime. Its units digit is 8.

M1-22

Mock Test 3 - Mathematics 1

Answer **D**

QUESTION 22

In the addition problem shown, m, n, p, and q represent positive digits. When the problem is completed correctly, the value of $m + n + p + q$ is (Canadian Math Competition) t is prime.

t is even.

t is divisible by 7.

)ne of its digits is 9.

information allows M

A 16

B 18

C 22

D 24

E 30

WORKED SOLUTION

From the units column, $3 + 2 + q$ ends in 2, so $q = 7$ with carry 1. From the tens column, $6 + p + 8 + 1$ ends in 4, so $p = 9$. From the hundreds column,

$n+7+5+2$ ends in 0, so $n=6$ with carry $m=2$. Therefore $m+n+p+q=2+6+9+7=24$.

M1-23 Mock Test 3 - Mathematics 1 Answer **C**

QUESTION 23

If two numbers x and y satisfy that $(x+1)^2+(y-5)^2=0$, what is $x+y$?

A −1

B 0

C 4

D 6

E 7

WORKED SOLUTION

A sum of squares is zero only if both squares are zero. Thus $x+1=0$ and $y-5=0$, so $x=-1, y=5$, and $x+y=4$.

M1-24 Mock Test 3 - Mathematics 1 Answer **D**

QUESTION 24

Abel tosses two coins and Ben tosses one coin. What is the probability that they get the equal number of heads?

Recall that

$$\text{Probability} = \frac{\text{Number of valid cases}}{\text{Number all possible cases}}.$$

A $\frac{1}{8}$

B $\frac{1}{4}$

C $\frac{1}{3}$

D $\frac{3}{8}$

E $\frac{2}{5}$

WORKED SOLUTION

There are 8 equally likely outcomes. Equal heads occurs when Abel gets 0 heads and Ben gets 0 heads, or when Abel gets 1 head and Ben gets 1 head: $1 + 2 = 3$ cases. Probability $= 3/8$.

M1-25 Mock Test 3 - Mathematics 1 Answer D

QUESTION 25

x, y, z and w are four numbers. If the average of the following three numbers $x - y,\ y - z,\ z - w$ is 10 and $w = 5$. What is x?

A 15

B 20

C 25

D 35

E there is no enough information

WORKED SOLUTION

The average of $x - y, y - z, z - w$ is 10, so their sum is 30. The sum telescopes: $(x - y) + (y - z) + (z - w) = x - w = 30$. Since $w = 5$, $x = 35$.

M1-26

Mock Test 3 - Mathematics 1

Answer A

QUESTION 26

What is the tens digit of 7^{2020}?

A 0

B 1

C 2

D 3

E 4

WORKED SOLUTION

The last two digits of powers of 7 cycle: $07, 49, 43, 01$. Since 2020 is divisible by 4, 7^{2020} ends in 01, so the tens digit is 0.

M1-27

Mock Test 3 - Mathematics 1

Answer **C**

QUESTION 27

Two trains 300 miles apart are traveling toward each other along the same track. One train goes 70 miles per hour; the other runs at 80 miles per hour. If both of them leave their stations at 7:00am, when will they meet?

A 8 : 00 am

B 8 : 30 am

C 9 : 00 am

D 9 : 30 am

E 10 : 00 am

WORKED SOLUTION

The trains approach each other at $70 + 80 = 150$ miles per hour. Time $= 300/150 = 2$ hours, so they meet at 9:00 am.

LEVEL 1 · STANDARD

Physics

27 QUESTIONS · COMPLETE SOLUTIONS

Engineering Mock Test 3 · preserved Physics module. Question wording, option order, answer and solution method are unchanged.

P-01

Mock Test 3 - Physics | Answer D

QUESTION 1

Which of the following statements are correct in relation to Newton's 3rd law?

- 1. For every action there is an equal and opposite reaction.
- 2. According to Newton's 3rd law, there are no isolated forces.
- 3. Rockets cannot accelerate in deep space because there is nothing to generate an equal and opposite force.

A. Only 1

B. Only 2

C. Only 3

D. 1 and 2

E. 2 and 3

F. 1 and 3

WORKED SOLUTION

Statement 1 is the common formulation of Newton's third law. Statement 2 presents a consequence of the application of Newton's third law.

Statement 3 is false: rockets can still accelerate because the products of burning fuel are ejected in the opposite direction from which the rocket needs to accelerate.

P-02

Mock Test 3 - Physics | Answer E

QUESTION 2

Which of the following statements are correct?

- 1. Positively charged objects have gained electrons.
- 2. Electrical charge in a circuit over a period of time can be calculated if the voltage and resistance are known.
- 3. Objects can be charged by friction.

A Only 1

B Only 2

C Only 3

D 1 and 2

E 2 and 3

F 1 and 3

G 1, 2 and 3

WORKED SOLUTION

Positively charged objects have lost electrons.

$$\text{Charge} = \text{Current} \times \text{Time} = \frac{\text{Voltage}}{\text{Resistance}} \times \text{Time}.$$

Objects can become charged by friction as electrons are transferred from one object to the other.

P-03

Mock Test 3 - Physics

Answer **B**

QUESTION 3

Which of the following statements is true?

- **A** The gravitational force between two objects is independent of their mass.
- **B** Each planet in the solar system exerts a gravitational force on the Earth.
- **C** For satellites in a geostationary orbit, acceleration due to gravity is equal and opposite to the lift from engines.
- **D** Two objects that are dropped from the Eiffel tower will always land on the ground at the same time if they have the same mass.
- **E** All of the above.
- **F** None of the above.

WORKED SOLUTION

Each body of mass exerts a gravitational force on another body with mass. This is true for all planets as well.

Gravitational force depends on the masses of both objects. Satellites stay in orbit due to centripetal force that acts tangentially to gravity (not because of the thrust from their engines). Two objects will only land at the same time if they also have the same shape or they are in a vacuum; otherwise air resistance gives different terminal velocities.

P-04

Mock Test 3 - Physics

Answer A

QUESTION 4

Which of the following best defines an electrical conductor?

- **A** Conductors allow electrical charge to pass through them easily because they contain mobile charge carriers and have low resistance.
- **B** Conductors are usually made from non-metals and they conduct electrical charge in multiple directions.
- **C** Conductors are usually made from metals and they conduct electrical charge in one fixed direction.
- **D** Conductors are usually made from non-metals and they conduct electrical charge in one fixed direction.
- **E** Conductors allow the passage of electrical charge with zero resistance because they contain freely mobile charged particles.
- **F** Conductors allow the passage of electrical charge with maximal resistance because they contain charged particles that are fixed and static.

WORKED SOLUTION

A conductor is a material that allows charge to move easily because it contains mobile charge carriers. Ordinary conductors have low resistance, not zero resistance.

P-05

Mock Test 3 - Physics

Answer E

QUESTION 5

An 800 kg compact car delivers 20% of its power output to its wheels. If the car has a mileage of 30 miles/gallon and travels at a speed of 60 miles/hour, how much power is delivered to the wheels? 1 gallon of petrol contains $9 \times 10^8\,\mathrm{J}$.

A 10 kW

B 20 kW

C 40 kW

D 50 kW

E 100 kW

WORKED SOLUTION

First, calculate the rate of petrol consumption:

$$\frac{\text{Speed}}{\text{Consumption}} = \frac{60\ \text{miles/ hour}}{30\ \text{miles/ gallon}} = 2\ \text{gallons/ hour}$$

Therefore, the total power is:

$$2\ \text{gallons} = 2 \times 9 \times 10^8 = 18 \times 10^8\ \mathrm{J}$$

$$1\ \text{hour} = 60 \times 60 = 3600\ \mathrm{s}$$

$$\text{Power} = \frac{\text{Energy}}{\text{Time}} = \frac{18 \times 10^8}{3600}$$

$$\mathrm{P} = \frac{18}{36} \times 10^6 = 5 \times 10^5\ \mathrm{W}$$

Since efficiency is 20%, the power delivered to the wheels is

$$5 \times 10^5 \times 0.2 = 10^5 \text{ W} = 100 \text{ kW}.$$

P-06

Mock Test 3 - Physics

Answer D

QUESTION 6

Which of the following statements about beta radiation are true?

- I. After a beta particle is emitted, the atomic mass number is unchanged.
- 2. Beta radiation can penetrate paper but not aluminium foil.
- 3. A beta particle is emitted from the nucleus of the atom when an electron changes into a neutron.

A I only

B 2 only

C I and 3

D I and 2

E 2 and 3

F I, 2 and 3

WORKED SOLUTION

Beta radiation is stopped by a few millimetres of aluminium, but not by paper. In β-radiation, a neutron changes into a proton plus an emitted electron, so the atomic mass number remains unchanged.

P-07

Mock Test 3 - Physics

Answer F

QUESTION 7

A car with a weight of 15,000 N is travelling at a speed of 15 $\mathrm{m\ s^{-1}}$ when it crashes into a wall and is brought to rest in 10 milliseconds. Calculate the average braking force exerted on the car by the wall. Take $g = 10\,\mathrm{m\ s^{-2}}$.

A $1.25 \times 10^4\,\mathrm{N}$

B $1.25 \times 10^5\,\mathrm{N}$

C $1.25 \times 10^6\,\mathrm{N}$

D $2.25 \times 10^4\,\mathrm{N}$

E $2.25 \times 10^5\,\mathrm{N}$

F $2.25 \times 10^6\,\mathrm{N}$

WORKED SOLUTION

Firstly, calculate the mass of the car:

$$\text{mass} = \frac{\text{Weight}}{g} = \frac{15000}{10} = 1500\ \text{kg}.$$

Then using $v = u + at$ where $v = 0\ \mathrm{m\ s^{-1}}$, $u = 15\ \mathrm{m\ s^{-1}}$ and $t = 10 \times 10^{-3}\ \mathrm{s} = 0.01\ \mathrm{s}$.

$$a = \frac{0 - 15}{0.01} = 1500\ \mathrm{m\ s^{-2}}.$$

$$F = ma = 1500 \times 1500 = 2\,250\,000\ \mathrm{N}.$$

P-08

Mock Test 3 - Physics

Answer E

QUESTION 8

Which of the following statements are correct?

- I. Electrical insulators are usually metals e.g. copper.
- 2. The flow of charge through electrical insulators is extremely low.
- 3. Electrical insulators can be charged by rubbing them together.

A Only 1

B Only 2

C Only 3

D I and 2

E 2 and 3

F I and 3

G I, 2 and 3

WORKED SOLUTION

Electrical insulators offer high resistance to the flow of charge. Insulators are usually non-metals; metals conduct charge very easily. Since charge does not flow easily to even out, they can be charged with friction.

P-09

Mock Test 3 - Physics

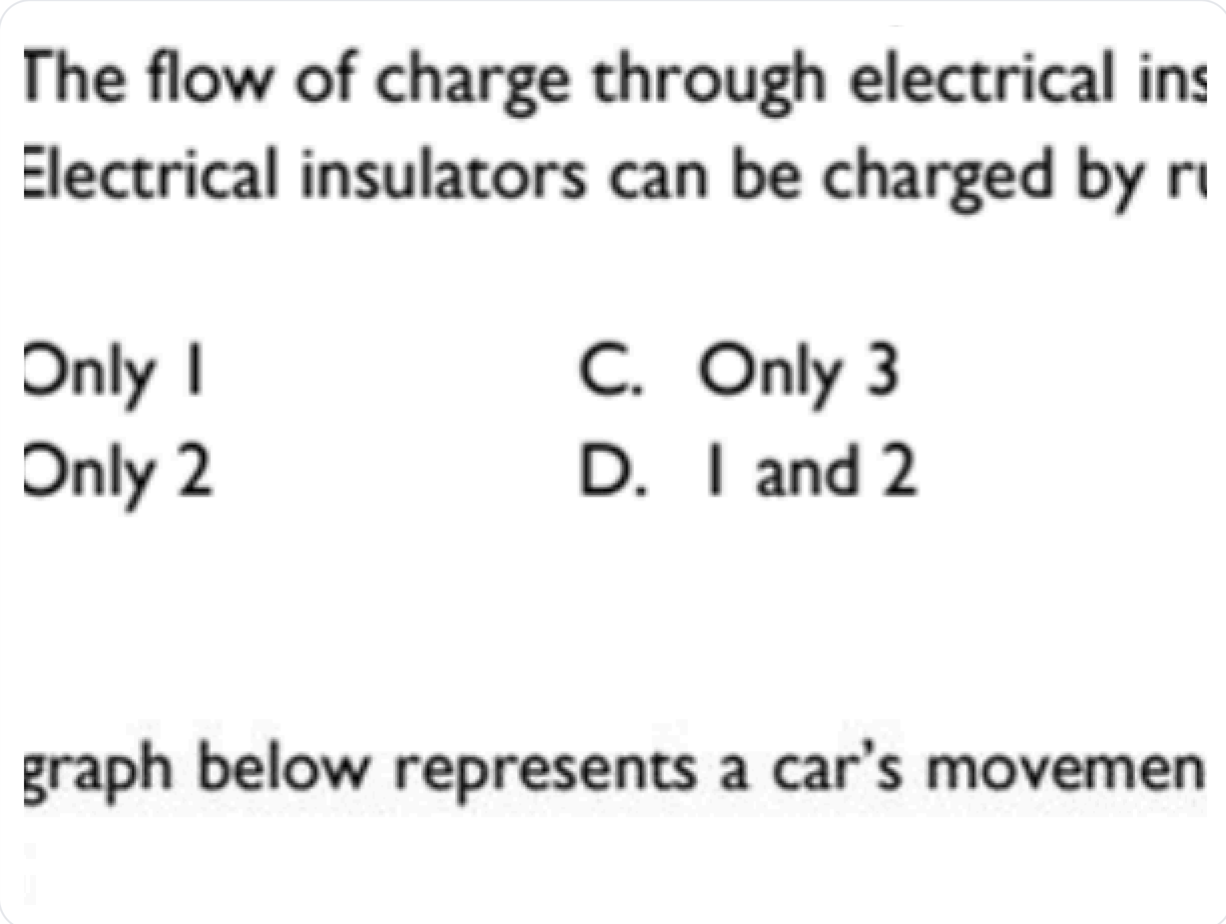
Answer A

QUESTION 9

The graph below represents a car's movement. At $t = 0$ the car's displacement was 0 m.

Which of the following statements are NOT true?

- 1. The car is reversing after $t = 30$.
- 2. The car moves with constant acceleration from $t = 0$ to $t = 10$.
- 3. The car moves with constant speed from $t = 10$ to $t = 30$.



The flow of charge through electrical ins
Electrical insulators can be charged by ru

Only I
Only 2
C. Only 3
D. I and 2

graph below represents a car's movemen

A 1 only

B 2 only

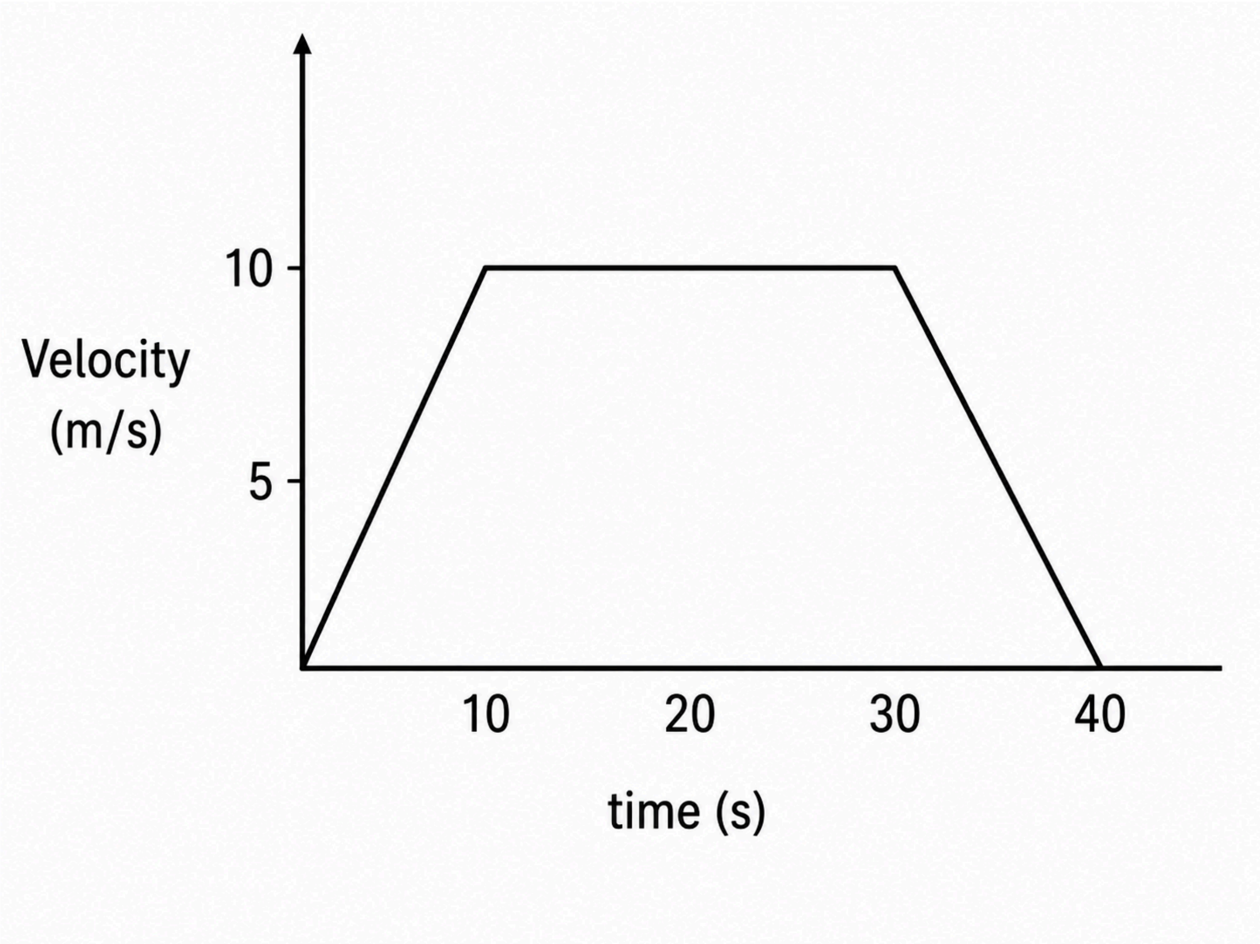
C 3 only

D 1 and 3

E 1 and 2

F 2 and 3

G 1, 2 and 3



WORKED SOLUTION

The car accelerates for the first 10 seconds at a constant rate and then decelerates after $t = 30$ seconds. It does not reverse, as the velocity is not negative. Therefore, only statement I is not true.

P-10

Mock Test 3 - Physics

Answer C

QUESTION 10

A 1,000 kg rocket is launched during a thunderstorm and reaches a constant velocity 30 seconds after launch. Suddenly, a strong gust of wind acts on it for 5 seconds with a force of 10,000 N in the direction of movement. What is the resulting change in velocity?

- **A** $0.5 \mathrm{~m~s}^{-1}$
- **B** $5 \mathrm{~m~s}^{-1}$
- **C** $50 \mathrm{~m~s}^{-1}$
- **D** $500 \mathrm{~m~s}^{-1}$
- **E** $5000 \mathrm{~m~s}^{-1}$
- **F** More information needed

WORKED SOLUTION

Using $F = ma$, where the unknown acceleration is $a = \dfrac{\Delta v}{\Delta t}$.

Hence: $\dfrac{F}{m} = \dfrac{\Delta v}{\Delta t}$.

Given $F = 10{,}000 \mathrm{~N}$, $m = 1{,}000 \mathrm{~kg}$ and $\Delta t = 5 \mathrm{~s}$.

So $\dfrac{10{,}000}{1{,}000} = \dfrac{\Delta v}{5}$.

Therefore $\Delta v = 10 \times 5 = 50 \mathrm{~m~s}^{-1}$.

P-11

Mock Test 3 - Physics

Answer D

QUESTION 11

A 0.5 tonne crane lifts a 0.01 tonne wardrobe by 100 cm in 5,000 milliseconds. Calculate the average power developed by the crane. Take $g = 10\ \mathrm{m\ s^{-2}}$.

- A 0.2 W
- B 2 W
- C 5 W
- D 20 W
- E 50 W
- F More information needed

WORKED SOLUTION

This question tests conversion to SI units and selection of relevant values.

0.01 tonnes = 10 kg; 100 cm = 1 m; 5,000 ms = 5 s.

$$P = \frac{\text{Work Done}}{t} = \frac{F \times d}{t}$$

The force is the weight of the wardrobe: $10 \times g = 10 \times 10 = 100\,\mathrm{N}$.

Thus, $P = \frac{100 \times 1}{5} = 20\,\mathrm{W}$.

P-12 Mock Test 3 - Physics Answer F

QUESTION 12

A 20 V battery is connected to a circuit consisting of a 1 Ω and 2 Ω resistor in parallel. Calculate the overall current of the circuit.

A 6.67 A

B 8 A

C 10 A

D 12 A

E 20 A

F 30 A

WORKED SOLUTION

Recall the resistance of a parallel circuit: $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots$.

Thus, $\frac{1}{R_T} = 1 + \frac{1}{2} = \frac{3}{2}$ and therefore $R = \frac{2}{3}\,\Omega$.

Using Ohm's Law: $I = \frac{V}{R} = \frac{20\text{ V}}{\frac{2}{3}\,\Omega} = 20 \times \frac{3}{2} = 30\text{ A}$.

P-13 Mock Test 3 - Physics Answer E

QUESTION 13

Which of the following statements is correct?

Assume the light ray crosses the boundary at a non-zero angle to the normal.

A The speed of light changes when it enters water.

B The speed of light changes when it leaves water.

C The direction of light changes when it enters water.

D The direction of light changes when it leaves water.

E All of the above.

F None of the above.

WORKED SOLUTION

At a non-zero angle to the normal, light changes speed when it passes between air and water and also changes direction due to refraction. Hence all four statements are true under the stated assumption.

P-14

Mock Test 3 - Physics

Answer **H**

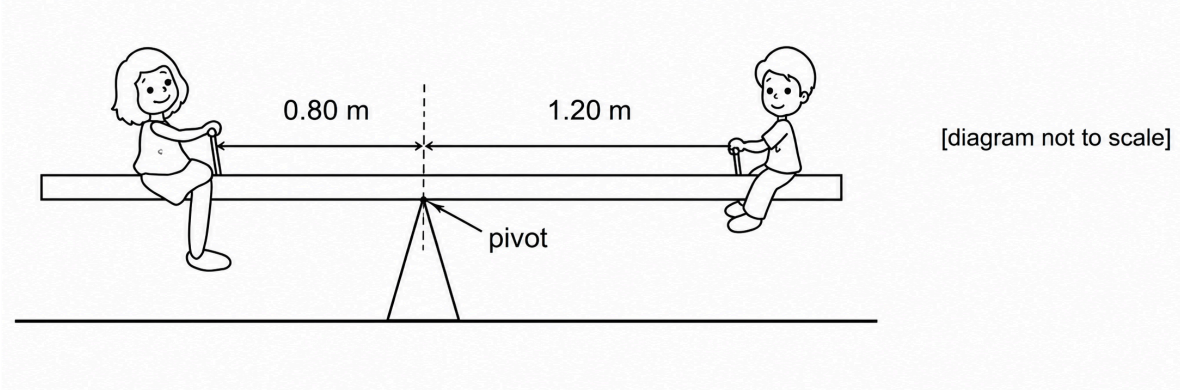
QUESTION 14

A plank of non-uniform density which has a mass of 15 kg is used to make a see-saw. A pivot is placed under the centre of the plank as shown on the diagram.

A boy of mass 35 kg sits at one end of the plank with his centre of gravity 1.20 m from the pivot. The see-saw balances when a woman of mass 60 kg sits on the plank on the other side of the pivot. Her centre of gravity is 0.80 m from the pivot.

Where is the centre of gravity of the plank and what is the magnitude of the force between the pivot and the plank?

(The gravitational field strength g is 10 N kg^-1.)



A 0.40 m on left of pivot; 100 N

B 0.40 m on left of pivot; 1100 N

C at the pivot; 100 N

D at the pivot; 1100 N

E 0.20 m on right of pivot; 100 N

F 0.20 m on right of pivot; 1100 N

G 0.40 m on right of pivot; 100 N

H 0.40 m on right of pivot; 1100 N

WORKED SOLUTION

Assume the centre of mass is at x metres to the right of the pivot.

Moments about pivot (CW positive):

$$35g(1.20) + Wx - 60g(0.8) = 0$$

$$Wx = 60g(0.8) - 35g(1.2)$$

$$Wx = 600(0.8) - 350(1.2) = 480 - 420 = 60$$

$$x = \frac{60}{W} = \frac{60}{150} = \frac{2}{5} = 0.40\,\text{m}$$

Newton's Second Law (vertical equilibrium on the plank):

$$R - W - 35g - 60g = 0$$

$$R = 150 + 350 + 600 = 1100\,\mathrm{N}$$

P-15

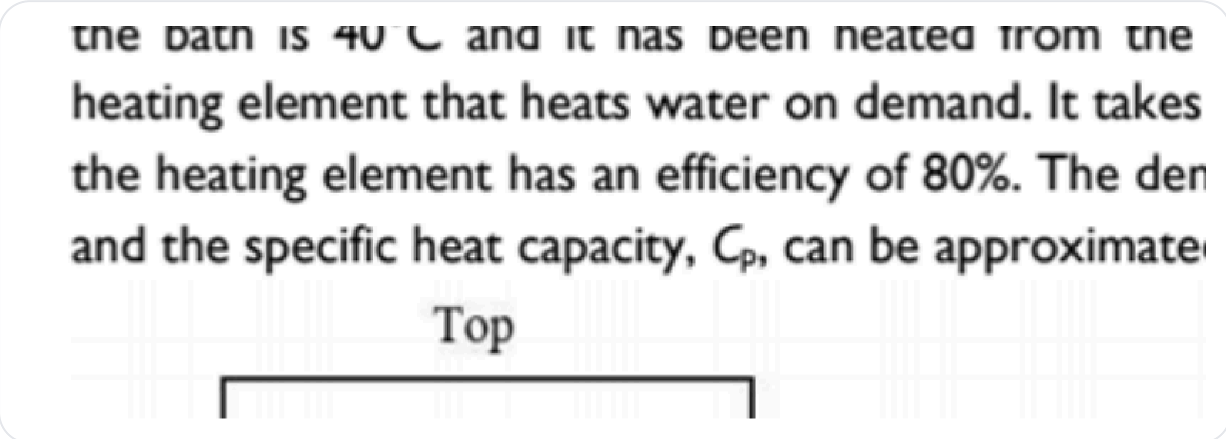
Mock Test 3 - Physics

Answer **C**

QUESTION 15

A bath that is 1.0 m by 1.5 m is filled with water to a depth of 30 cm. The water in the bath is 40°C and it has been heated from the mains water at 20°C using a heating element that heats water on demand. It takes 3 minutes to run the bath and the heating element has an efficiency of 80%. The density of water, ρ, is 1000 kg m^{-3}, and the specific heat capacity, C_p, can be approximated as 4000 J kg^{-1} K^{-1}.

What power is used by the heating element?

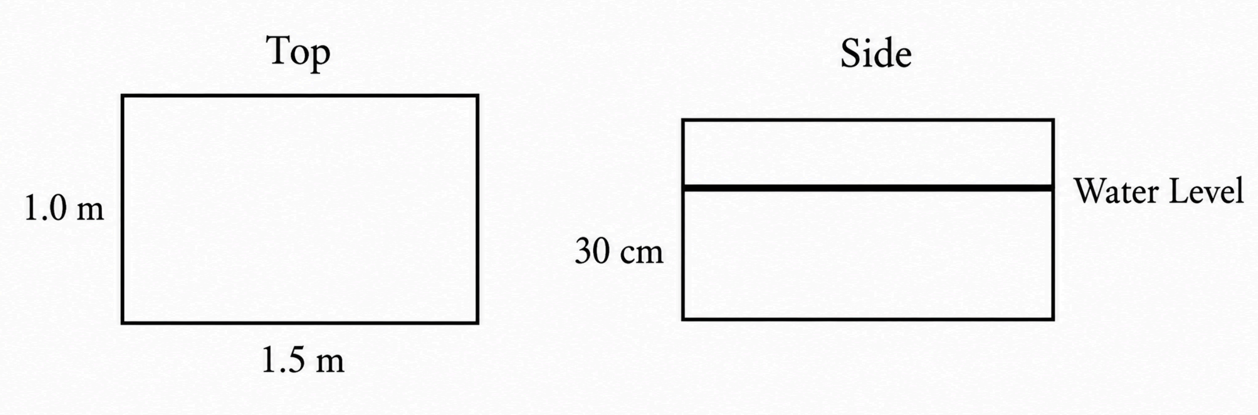


- **A** 2500 kW
- **B** 2000 kW
- **C** 250 kW
- **D** 200 kW
- **E** 100 kW

F 360 kW

G 3600 kW

H 1000 kW



WORKED SOLUTION

Answer: C

Work out how much energy is used to heat the water, then divide by the time taken to fill the bath to find the power used without accounting for losses. Then take account of the heating element efficiency to find the total power used.

Find the mass of water:

$m = \rho V$

$V = 1 \times 1.5 \times 0.3 = 0.45\ m^3$

$\therefore m = 1000 \times 0.45 = 450\ kg$

Energy: $E = m c_p \Delta T$

$E = 450 \times 4000 \times 20 = 3.6 \times 10^7\ J$

$$P = \frac{\Delta E}{\Delta t} = \frac{3.6 \times 10^7}{60 \times 3} = 2 \times 10^5\ W = 200\ kW$$

$P = \text{Efficiency} \times P_{tot}$

$$\therefore P_{tot} = \frac{P}{\text{Efficiency}} = \frac{200}{0.8} = 250\ kW$$

P-16

Mock Test 3 - Physics

Answer F

QUESTION 16

A 5 kW power source is used to generate a current through a circuit with a resistance of $100\ \Omega$. The charge of an electron, e, is 1.6×10^{-19} C. How many electrons pass through the circuit in 16 seconds?

- **A** 4×10^{21}
- **B** 5×10^{20}
- **C** $4\sqrt{3} \times 10^{19}$
- **D** 5×10^{18}
- **E** $5\sqrt{2} \times 10^{21}$
- **F** $5\sqrt{2} \times 10^{20}$
- **G** $8\sqrt{2} \times 10^{22}$
- **H** $8\sqrt{2} \times 10^{21}$

WORKED SOLUTION

Answer: F

The current is the flow of charge in the circuit. Work out the current and use it to find the total charge and consequently the total number of electrons that flow in a given time.

$$P = I^2R$$

$$I^2 = \frac{P}{R} = \frac{5000}{100} = 50\ A^2$$

$I = \sqrt{50} = 5\sqrt{2}\ A$

Remember where Q is charge: $I = \dfrac{\Delta Q}{\Delta t}$

$\therefore \Delta Q = \Delta t \times I = 16 \times 5\sqrt{2}\ C$

Number of electrons $= \dfrac{\Delta Q}{e}$

$= \dfrac{16 \times 5\sqrt{2}}{1.6 \times 10^{-19}} = \dfrac{16 \times 5\sqrt{2}}{16 \times 10^{-20}} = 5\sqrt{2} \times 10^{20}$

P-17

Mock Test 3 - Physics

Answer A

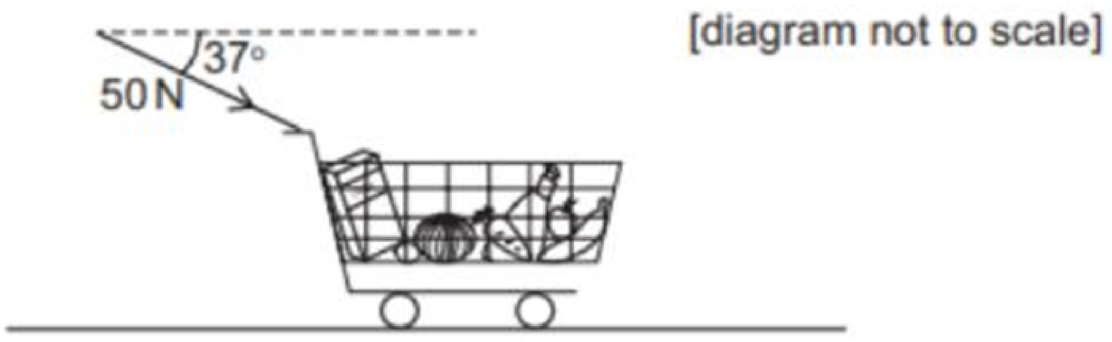
QUESTION 17

A shopper pushes a supermarket trolley a distance of 15 m in a straight line across a level, horizontal surface. The shopper applies a constant force of 50 N at an angle of 37° below the horizontal. The total weight of the trolley and its contents is 350 N.

[diagram not to scale]

What is the magnitude of the total vertical force that the surface exerts on the trolley and how much work is done by the pushing force?

(You may use the approximations $\sin 37^\circ = 0.60$; $\cos 37^\circ = 0.80$.)



A vertical force 380 N; work done 600 J

B vertical force 380 N; work done 750 J

C vertical force 390 N; work done 450 J

D vertical force 390 N; work done 750 J

WORKED SOLUTION

Question 17: A

Newtons Second law upwards on trolley:

$$\mathrm{R} - 350 - 50\sin(37) = 0$$
$$\mathrm{R} - 350 - 50(0.6) = 0$$
$$\mathrm{R} = 350 + 30 = 380\,\mathrm{N}$$

Work done = Force × Distance:

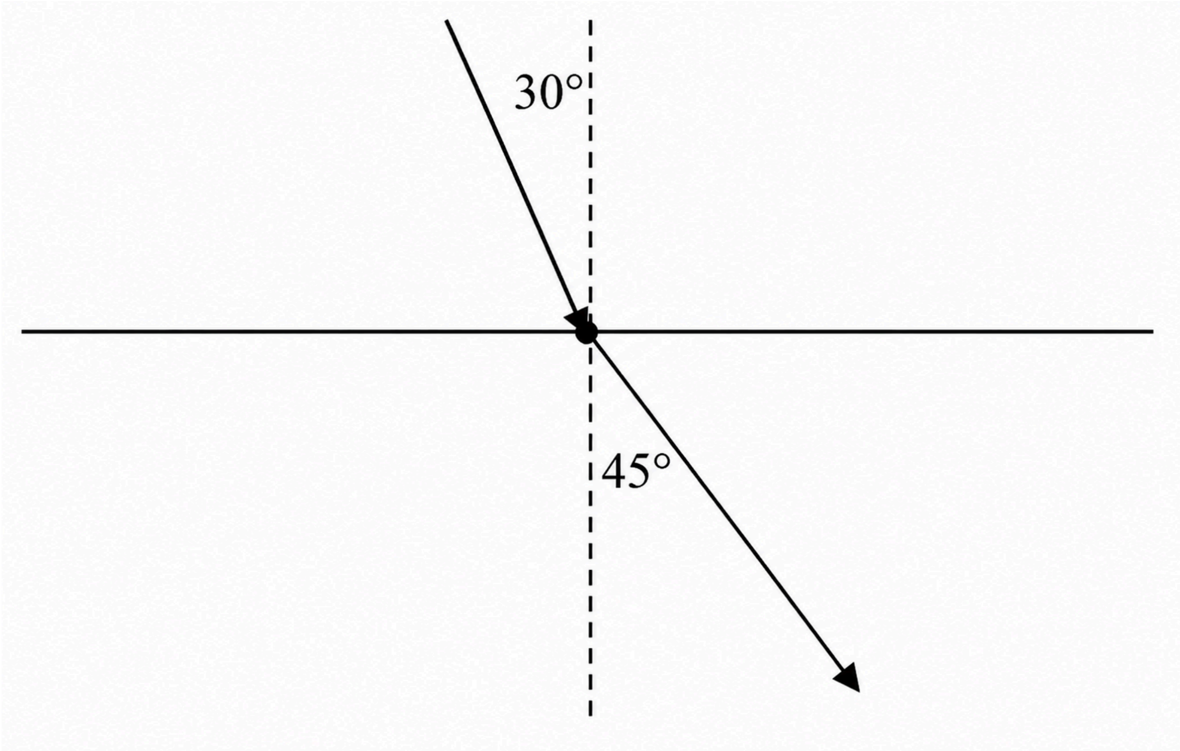
$$\mathrm{W} = 50\cos(37) \times 15$$
$$\mathrm{W} = 50(0.80) \times 15$$
$$\mathrm{W} = 40 \times 15 = 600\,\mathrm{J}$$

P-18 Mock Test 3 - Physics Answer **H**

QUESTION 18

A wave approaches the boundary between two mediums at an angle of 30° to the normal. Upon entering the second medium it is refracted such that it now travels at 45° to the normal. The wave travels at $5\,\mathrm{m\,s^{-1}}$ in the first medium.

What velocity does the wave travel at in the second medium?



A $\frac{8}{\sqrt{3}}$ m s^{-1}

B $8\sqrt{3}$ m s^{-1}

C $\frac{8}{\sqrt{2}}$ m s^{-1}

D $10\sqrt{3}$ m s^{-1}

E $\frac{5}{2}$ m s^{-1}

F $10\sqrt{2}$ m s^{-1}

G $\frac{5}{\sqrt{2}}$ m s^{-1}

H $5\sqrt{2}$ m s^{-1}

WORKED SOLUTION

Question 18: H

$$n = \frac{v_1}{v_2} = \frac{\sin\theta_1}{\sin\theta_2}$$

$$n = \frac{\sin 30}{\sin 45}$$

Remember: $\sin 30 = \frac{1}{2}, \ \sin 45 = \frac{1}{\sqrt{2}}$

$$\therefore \ n = \frac{1}{2} \div \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

$$v_2 = \frac{v_1}{n} = 5 \div \frac{\sqrt{2}}{2} = \frac{10}{\sqrt{2}}$$

$$\therefore \ v_2 = 5\sqrt{2}\ \mathrm{m\ s^{-1}}$$

P-19

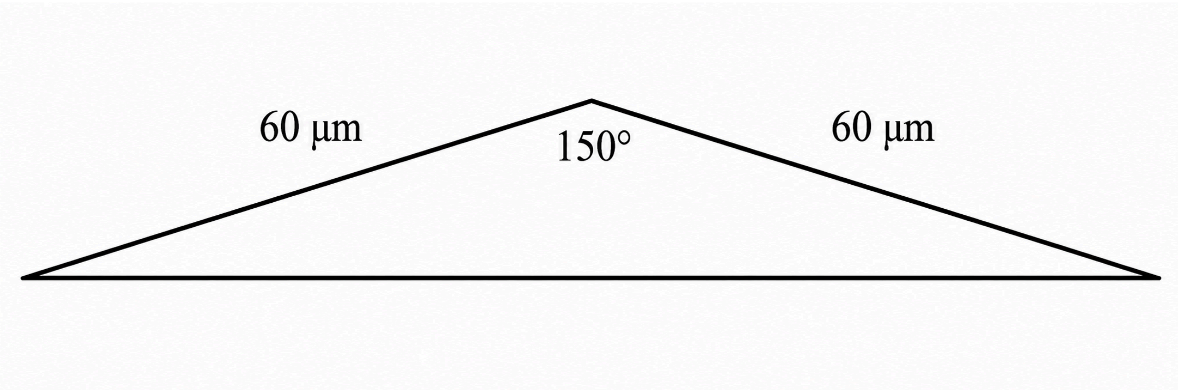
Mock Test 3 - Physics

Answer D

QUESTION 19

A strip of conductive material is plated onto a silicon surface to allow an electronic signal to be transmitted. A cross section of the plated material is given below, and the resistance is measured over a 5 cm length of the strip to be 1.0 Ω.

What is the resistivity of the material?



A $6.0 \times 10^{-7}\,\Omega\mathrm{m}$

B $3.6 \times 10^{-9}\,\Omega\mathrm{m}$

C $2.4 \times 10^{-8}\,\Omega\,m$

D $1.8 \times 10^{-8}\,\Omega\,m$

E $2.0 \times 10^{-4}\,\Omega\,m$

F $3.6 \times 10^{-10}\,\Omega\,m$

G $5.4 \times 10^{-11}\,\Omega\,m$

H $4.2 \times 10^{-11}\,\Omega\,m$

WORKED SOLUTION

Work out the area of the cross section of the strip and use that to find the resistivity.

resistivity, $\rho = \text{resistance} \times \dfrac{\text{cross sectional area}}{\text{length}} = \dfrac{RA}{l}$

Recall area of triangle: $A = \frac{1}{2}ab\sin C$.

$A = \frac{1}{2}(6 \times 10^{-5})^2 \sin(150°)$

Remember $\sin\theta$ is symmetrical about $90°$ so $\sin(150°) = \sin(30°) = \frac{1}{2}$.

$\therefore A = \frac{1}{2}(6 \times 10^{-5})^2 \sin(30°)$

$A = \frac{1}{2}(6 \times 10^{-5})^2 \times \frac{1}{2}$

$A = \frac{1}{4} \times 36 \times 10^{-10} = 9 \times 10^{-10}\,m^2$

$\rho = \dfrac{RA}{l} = \dfrac{1 \times 9 \times 10^{-10}}{0.05} = 9 \times 10^{-10} \times 20 = 1.8 \times 10^{-8}\,\Omega\,m$

Note: This is approximately the resistivity of copper.

P-20

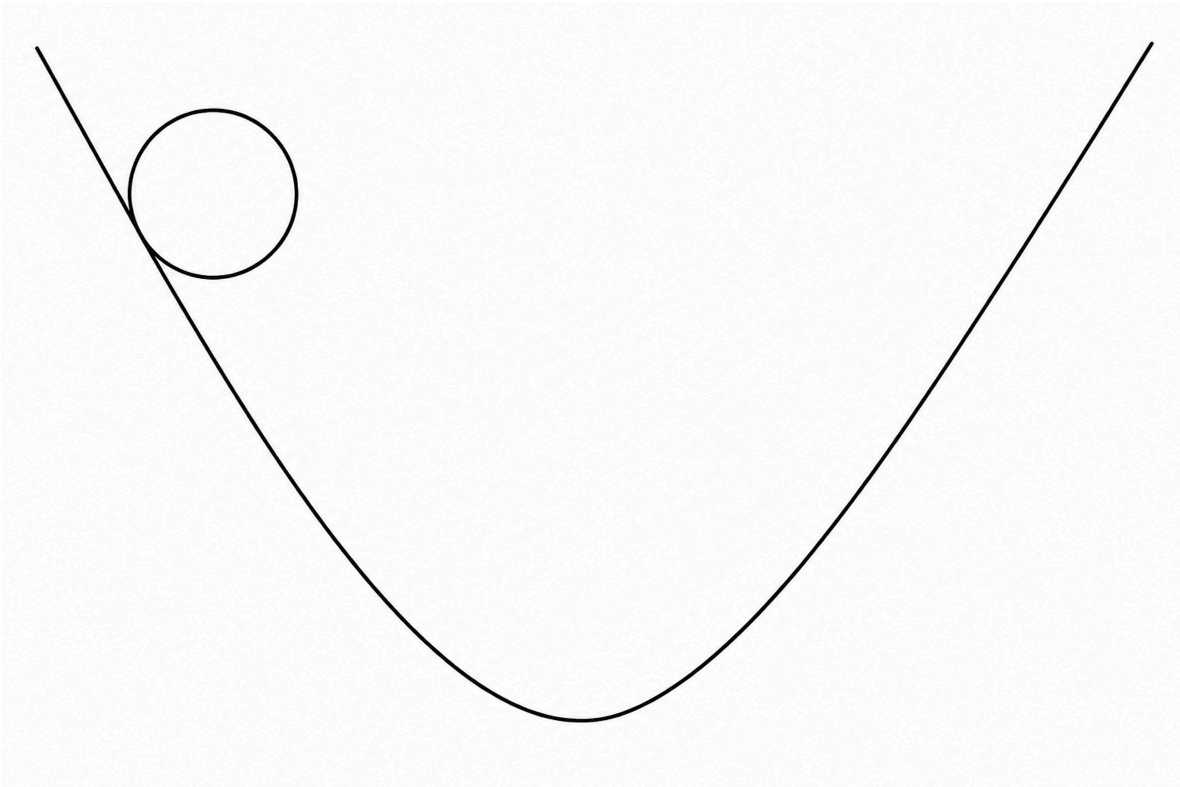
Mock Test 3 - Physics

Answer A

QUESTION 20

A ball rolls down and up a dip which can have its side profile modelled by the equation $y = 2x^2$. The point $(0, 0)$ corresponds to the bottom of the dip, and y is the height of the surface at horizontal distance x away from this centre.

What is the angle of the contact force acting on the ball when it is 3 m away from the lowest point horizontally? (Give the smallest angle between the direction of the force and the horizontal.)



- **A** $\tan^{-1}(\frac{1}{12})$
- **B** $\tan^{-1}(12)$
- **C** $\tan^{-1}(8)$
- **D** $\tan^{-1}(\frac{1}{8})$

E $\sin^{-1}(\frac{1}{12})$

F $\sin^{-1}(12)$

G $\sin^{-1}(8)$

H $\sin^{-1}(\frac{1}{8})$

WORKED SOLUTION

Question 20: A

You can evaluate the angle by looking at either $x = -3$ or $x = 3$ as the dip is symmetrical. Use differentiation to find the gradient at either point then use the gradient to work out the angle between the force and horizontal.

Find gradient at $x = 3$:

$$y = 2x^2$$
$$\frac{dy}{dx} = 4x$$
$$\Rightarrow \frac{dy}{dx} = 4 \times 3 = 12$$

Direction of the force is normal to the surface the ball is rolling on:

$$m' = -\frac{1}{m} = -\frac{1}{12}$$

Find the angle with the horizontal of this gradient:

$$\tan\theta = \frac{1}{12}$$

$$\theta = \tan^{-1}(\frac{1}{12})$$

Note: This is approximately $5°$ but trig function is given as the paper is non-calculator.

P-21

Mock Test 3 - Physics

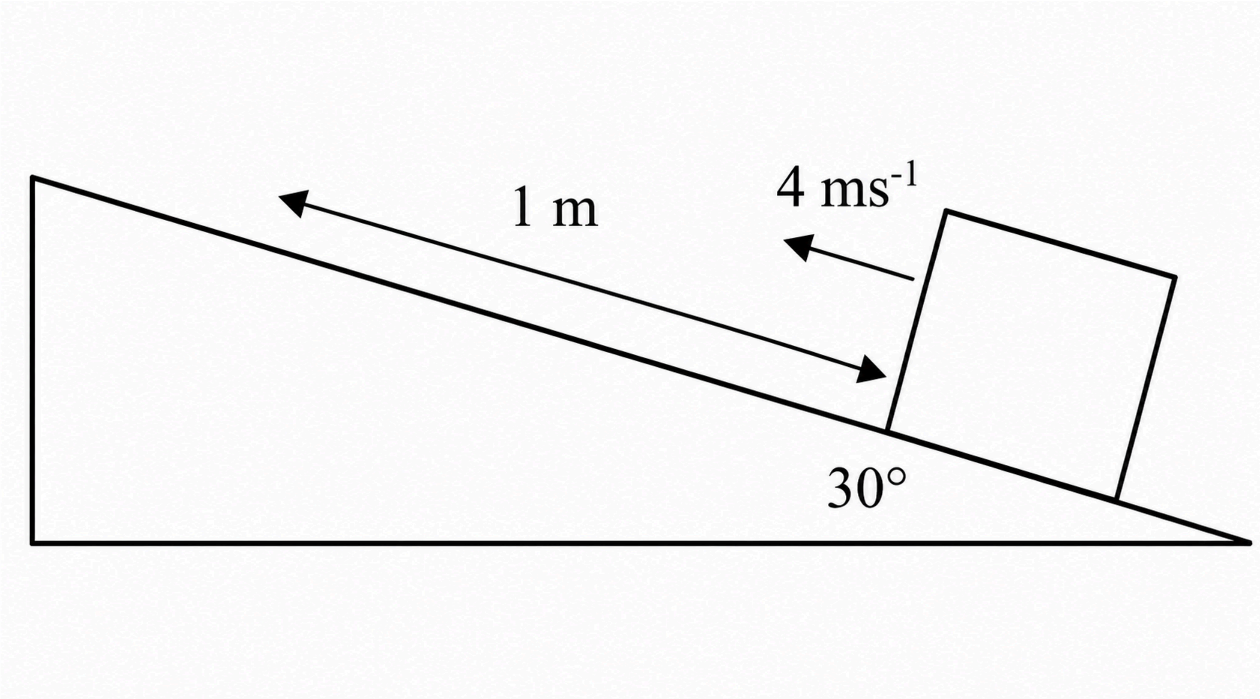
Answer D

QUESTION 21

A block of mass 2 kg sliding up a slope has an initial velocity of $4\,\mathrm{m\,s^{-1}}$. The slope is angled at $30°$ to the horizontal and the block moves 1 m before it stops. (Take $g = 10\,\mathrm{N\,kg^{-1}}$).

What is the average frictional force on the block as it moves up the slope?

- A 2 N
- B 1 N
- C 3 N
- D 6 N
- E 4 N
- F 5 N
- G 0 N
- H 7 N



WORKED SOLUTION

Question 21: D

Do an energy balance for the conversion of kinetic energy into gravitational potential energy and work done against friction.

Change in height:

$$h = s\sin\theta = 1\sin 30 = 10 \times \frac{1}{2} = 0.5\,\mathrm{m}$$

Energy balance:

$$KE = GPE + WD$$

$$\frac{1}{2}mv^2 = mgh + F_f s$$

$$\frac{1}{2} \times 2 \times 4^2 = 2 \times 10 \times 0.5 + F_f \times 1$$

$$16 = 10 + F_f$$

$$F_f = 6\,\mathrm{N}$$

P-22

Mock Test 3 - Physics

Answer F

QUESTION 22

An object is pushed with a force of 3 N for 5 s and then a second force of 4 N for 4 s. The forces act in the same direction and push an object of mass 5 kg. At the end of the 9 s period, what is the total change in momentum?

A $16\,\mathrm{kg\,m\,s^{-1}}$

B $32\,\mathrm{kg\,m\,s^{-1}}$

C $8\,\mathrm{kg\,m\,s^{-1}}$

D $12\,\mathrm{kg\,m\,s^{-1}}$

E $20\,\mathrm{kg\,m\,s^{-1}}$

F $31\,\mathrm{kg\,m\,s^{-1}}$

G $4\,\mathrm{kg\,m\,s^{-1}}$

H $24\,\mathrm{kg\,m\,s^{-1}}$

WORKED SOLUTION

The impulse equals the change in momentum. The two impulses act in the same direction, so

$\Delta p = 3 \times 5 + 4 \times 4 = 15 + 16 = 31\,\mathrm{kg\,m\,s^{-1}}$.

P-23

Mock Test 3 - Physics

Answer E

QUESTION 23

Light travels at a speed of $2.5 \times 10^{8}\,\mathrm{m\,s^{-1}}$ in a material. What is the critical angle for total internal reflection when light approaches an air boundary from within the material? (The speed of light in air is $3 \times 10^{8}\,\mathrm{m\,s^{-1}}$).

A 30°

B 45°

C 60°

D $\sin^{-1}(\frac{4}{5})$

E $\sin^{-1}(\frac{5}{6})$

F $\sin^{-1}(\frac{3}{5})$

G $\sin^{-1}(\frac{2}{3})$

H $\sin^{-1}(\frac{4}{7})$

WORKED SOLUTION

Find the refractive index and then the critical angle.

$$n = \frac{c}{v}$$

$$n = \frac{3 \times 10^{8}}{2.5 \times 10^{8}} = \frac{6}{5} = 1.2$$

For the critical angle, $\sin C = \dfrac{1}{n}$.

$$\sin C = \frac{1}{1.2} = \frac{5}{6}$$

$$C = \sin^{-1}(\frac{5}{6}).$$

P-24

Mock Test 3 - Physics

Answer B

QUESTION 24

A spring is used to support a mass of 2 kg. The spring's spring constant, k, is $50\,\text{N m}^{-1}$. How much elastic potential energy is stored in the spring due to the mass it is supporting? (Take $g = 10\,\text{N kg}^{-1}$.)

- **A** 1 J
- **B** 4 J
- **C** 20 J
- **D** 25 J
- **E** 8 J
- **F** 15 J
- **G** 14 J
- **H** 30 J

WORKED SOLUTION

Work out the extension using Hooke's law and then the elastic potential energy.

$$W = F = kx$$

$$mg = kx$$

$$2 \times 10 = 50x \Rightarrow x = \frac{20}{50} = 0.4\,m$$

$$E = \tfrac{1}{2}Fx = \tfrac{1}{2}mgx$$

$$E = \tfrac{1}{2} \times 2 \times 10 \times 0.4 = 4\,J$$

P-25

Mock Test 3 - Physics

Answer H

QUESTION 25

A charge of 500 C is unloaded over a 30 s period. The voltage in the circuit is 90 V. What is the average power obtained?

- A 600 W
- B 450
- C 250 W
- D 300 W
- E 100 W
- F 1800 W
- G 1000 W
- H 1500 W

WORKED SOLUTION

Find the current from charge and time, then compute power.

$$I = \frac{\Delta Q}{\Delta t}$$

$$I = \frac{500}{30} = \frac{50}{3} \text{ A}$$

$$P = IV$$

$$\therefore P = \frac{50}{3} \times 90 = 1500 \text{ W}$$

P-26

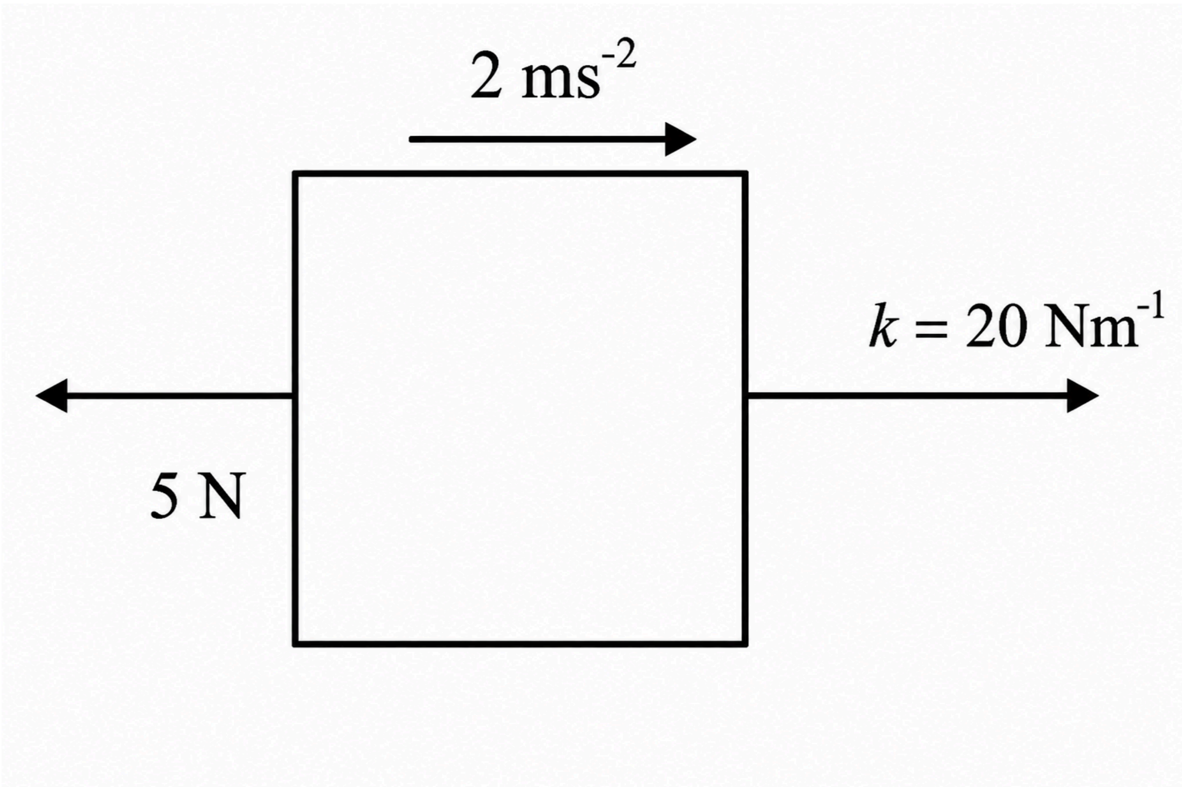
Mock Test 3 - Physics

Answer G

QUESTION 26

A block of mass 4 kg is pulled with a spring of spring constant, $k = 20 \text{ N m}^{-1}$. The acceleration of the block is 2 m s^{-2} and the force due to friction, F_f, is 5 N.

What is the extension in the spring?



A 22 cm

B 35 cm

C 8 cm

D 12 cm

E 5 cm

F 40 cm

G 65 cm

H 6 cm

WORKED SOLUTION

Resolve horizontally. The spring force must both overcome friction and accelerate the block:

$F_s - 5 = 4 \times 2$, so $F_s = 13\,\mathrm{N}$.

Using Hooke's law, $F_s = kx$, so $13 = 20x$. Hence $x = 0.65\,\mathrm{m} = 65\,\mathrm{cm}$.

P-27 Mock Test 3 - Physics Answer **C**

QUESTION 27

A cylindrical wire is made of a material with a resistivity of $8 \times 10^{-8}\,\Omega\mathrm{m}$. It has a radius of 0.5 mm, and a length of 1 km. If the voltage in the wire is 400 V, what is the current?

A $\frac{10\pi}{3}$ A

B $\frac{2\pi}{5}$ A

C $\frac{5\pi}{4}$ A

D $\frac{6\pi}{25}$ A

E $\frac{25\pi}{3}$ A

F $\frac{13\pi}{4}$ A

G $\frac{17\pi}{4}$ A

H $\frac{11\pi}{5}$ A

WORKED SOLUTION

Question 27: C

Find the resistance using the resistivity and dimensions of the wire. Then solve with $V = IR$.

resistivity, $\rho = \frac{RA}{l}$

$\therefore R = \frac{\rho l}{A}$

$A = \pi \times (0.5 \times 10^{-3})^2 = \frac{\pi}{4} \times 10^{-6} m^2$

$R = \frac{8 \times 10^{-8} \times 1000}{\frac{\pi}{4} \times 10^{-6}} = \frac{320}{\pi} \Omega$

$I = \frac{V}{R}$

$\therefore I = 400 \div \frac{320}{\pi} = \frac{5\pi}{4} A$

LEVEL 1 · STANDARD

Biology

27 QUESTIONS · COMPLETE SOLUTIONS

Preserved Biology Standard source set. Question wording, option order, answer and solution method are unchanged.

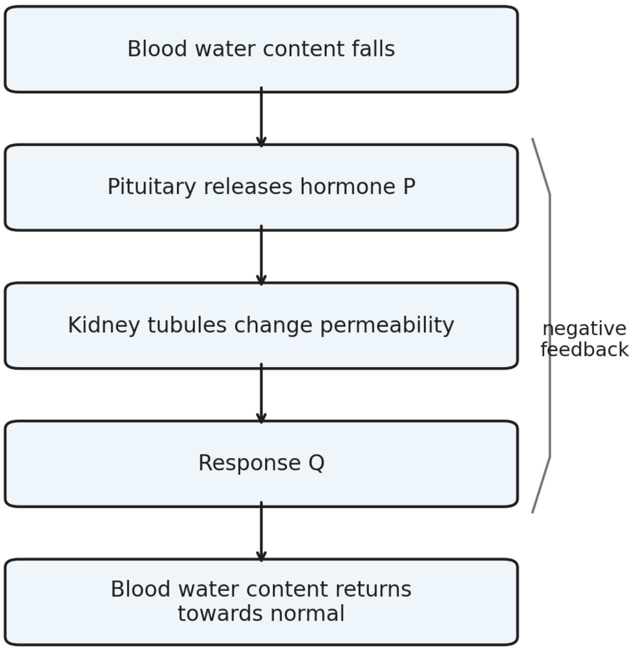
B-01

Homeostasis | Answer D

QUESTION 1

A person loses water by sweating for a prolonged period.

The diagram summarises part of the homeostatic response.



A insulin; more water reabsorbed by the kidneys

B glucagon; less water reabsorbed by the kidneys

C ADH; less water reabsorbed and a larger volume of urine produced

D ADH; more water reabsorbed and a smaller volume of urine produced

E adrenaline; less water reabsorbed and more concentrated urine

WORKED SOLUTION

Low blood water content causes increased release of ADH from the pituitary gland.

ADH increases the permeability of parts of the kidney tubule, so more water is reabsorbed into the blood. Less water is therefore lost in urine, giving a smaller volume of more concentrated urine.

Hence the correct answer is **D**.

B-02 Cell structure Answer **C**

QUESTION 2

A cell has:

- a cell membrane;
- cytoplasm;
- a cell wall;
- circular chromosomal DNA;
- several small plasmids;
- no mitochondria.

Which statement about the cell is correct?

A It must contain chloroplasts.

B It cannot carry genetic information.

C A plasmid may carry a gene for antibiotic resistance.

D It is a eukaryotic cell.

E It cannot carry out respiration.

WORKED SOLUTION

The combination of circular chromosomal DNA, plasmids and absence of mitochondria is characteristic of a bacterial cell.

Plasmids can carry additional genes, including genes that confer antibiotic resistance. Bacteria can still carry out respiration even though they do not possess mitochondria.

Therefore the correct answer is **C**.

B-03 DNA and base composition | Answer **C**

QUESTION 3

A double-stranded DNA molecule is **300 base pairs** long.

Adenine accounts for 36% of all bases in the molecule.

How many guanine bases are present?

A 42

B 72

C 84

D 108

E 168

WORKED SOLUTION

There are

$$300 \times 2 = 600$$

bases in total.

In double-stranded DNA, adenine and thymine occur in equal proportions. Therefore

$$A + T = 36\% + 36\% = 72\%.$$

The remaining 28% consists equally of cytosine and guanine, so guanine makes up 14%.

$$0.14 \times 600 = 84.$$

Thus the correct answer is **C**.

B-04

Natural selection

Answer G

QUESTION 4

A population of beetles contains both pale and dark individuals.

Before an industrial development, most tree bark in the beetles' habitat is pale. After many years of soot deposition, most bark becomes much darker. The frequency of an allele associated with dark colour subsequently increases.

Which statements could explain the change?

1. Soot caused pale beetles to mutate specifically into dark beetles.
2. Dark beetles may have become less visible to predators.
3. If colour is heritable, dark beetles surviving and reproducing more successfully could increase the frequency of the dark allele.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Mutation is not directed by environmental need, so statement 1 is incorrect.

Dark beetles could be better camouflaged on darkened bark, increasing their chance of survival. If the difference is heritable, greater reproductive success of dark beetles can increase the frequency of the associated allele.

Statements 2 and 3 are therefore correct, giving **G**.

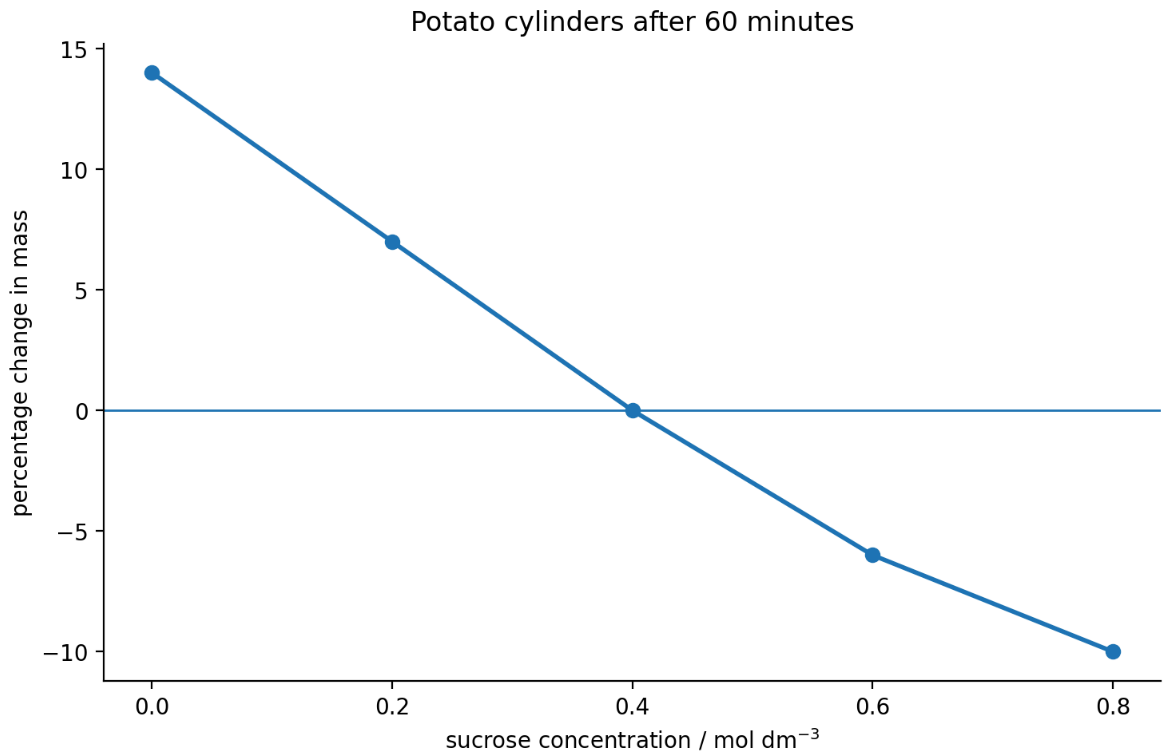
B-05

Osmosis | Answer **C**

QUESTION 5

Identical potato cylinders were placed for 60 minutes in different sucrose solutions.

Which row is correct?



	concentration giving approximately no net change in mass / mol dm^{-3}	net movement of water at 0.20 mol dm^{-3}
A	0.20	into the potato cells
B	0.40	out of the potato cells
C	0.40	into the potato cells
D	0.60	into the potato cells
E	0.40	no net movement

A row A

B row B

C row C

D row D

E row E

WORKED SOLUTION

At approximately 0.40 mol dm^{-3}, the potato cylinders show no percentage change in mass, indicating no net movement of water.

At 0.20 mol dm^{-3}, the cylinders gain mass, so there has been a net movement of water into the potato cells by osmosis.

Therefore row **C** is correct.

B-06 Enzymes Answer **F**

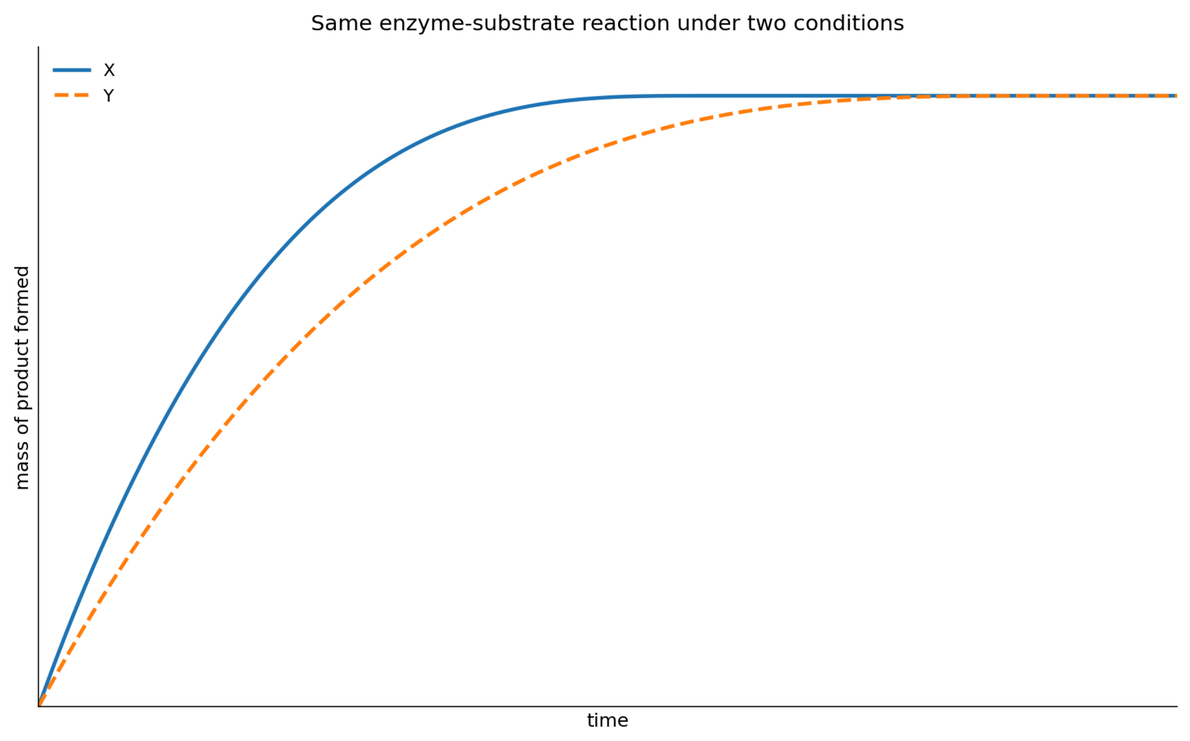
QUESTION 6

The graph shows the mass of product produced during the same enzyme-controlled reaction under two different conditions, X and Y.

The graph shows both reactions until no further product is formed.

Which changes **could** explain the difference between X and Y?

1. X contains a greater concentration of enzyme than Y.
2. X contains a greater initial amount of substrate than Y.
3. The temperature in X is closer to the enzyme's optimum than the temperature in Y, with neither temperature causing denaturation.



A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Both curves reach the same final mass of product, so the graph does not support X having a greater initial amount of substrate than Y.

A greater enzyme concentration can increase the rate of reaction without changing the total amount of substrate ultimately converted. A temperature closer to the enzyme's optimum can also increase the rate, provided neither temperature causes denaturation.

Therefore statements 1 and 3 could explain the graph: **F**.

B-07 Cell division Answer **E**

QUESTION 7

A human body cell contains 46 chromosomes and $6\ \mathrm{pg}$ of DNA before DNA replication.

Which row correctly describes the same cell immediately before mitosis and one daughter cell after mitosis?

	immediately before mitosis: chromosomes	immediately before mitosis: DNA / pg	one daughter cell after mitosis: chromosomes	one daughter cell after mitosis: DNA / pg
A	23	12	23	6
B	46	6	46	6
C	92	12	46	6
D	46	12	23	6
E	46	12	46	6

A row A

B row B

C row C

D row D

E row E

WORKED SOLUTION

DNA replication doubles the amount of DNA from 6 pg to 12 pg, but the chromosome number is still counted as 46 because each chromosome consists of two sister chromatids.

After mitosis, each daughter cell has the normal diploid chromosome number and half the replicated DNA:

$$46 \text{ chromosomes}, \qquad 6 \text{ pg DNA}.$$

So row **E** is correct.

B-08

Stem cells

Answer F

QUESTION 8

Which statements about an adult stem cell from human bone marrow are correct?

1. It may divide by mitosis to produce more stem cells.
2. It is totipotent and can form an entire new human organism.
3. It can differentiate into a limited range of specialised blood cells.

- **A** none
- **B** 1 only
- **C** 2 only
- **D** 3 only
- **E** 1 and 2 only
- **F** 1 and 3 only
- **G** 2 and 3 only
- **H** 1, 2 and 3

WORKED SOLUTION

Adult stem cells can divide by mitosis and self-renew. Bone-marrow stem cells are multipotent: they can form a restricted range of specialised blood cells.

They are not totipotent and cannot normally form an entire new organism.

Therefore statements 1 and 3 only are correct: **F**.

B-09

Inheritance and pedigrees

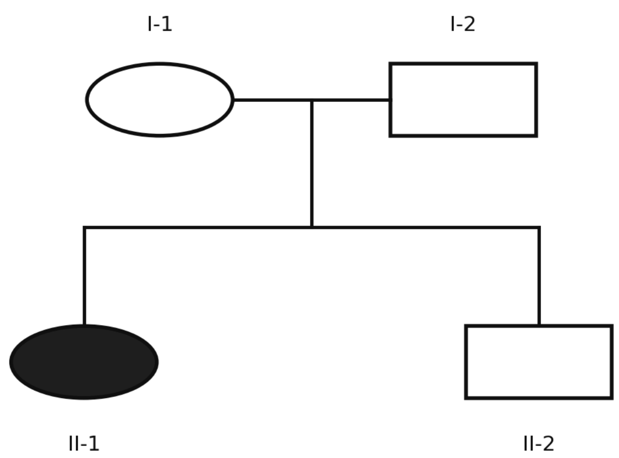
Answer C

QUESTION 9

The condition shown in the pedigree is caused by an **autosomal recessive allele**.

Individual II-2 is unaffected.

If II-2 later has a child with a woman who is affected by the condition, what is the probability that their child will be affected?



Filled symbol = affected by an autosomal recessive condition

A $\frac{1}{6}$

B $\frac{1}{4}$

C $\frac{1}{3}$

D $\frac{1}{2}$

E $\frac{2}{3}$

WORKED SOLUTION

Because two unaffected parents produced an affected child with a recessive condition, both parents must be heterozygous:

$$\mathrm{Aa} \times \mathrm{Aa}.$$

Given that II-2 is unaffected, the possible genotypes are $\mathrm{AA}, \mathrm{Aa}, \mathrm{Aa}$. Therefore

$$P\,(\text{II-2 is Aa}) = \frac{2}{3}.$$

The affected woman is aa. If II-2 is a carrier, the probability that their child is affected is $\frac{1}{2}$.

Hence

$$\frac{2}{3} \times \frac{1}{2} = \boxed{\frac{1}{3}}.$$

The correct answer is **C**.

B-10

Monohybrid inheritance

Answer **D**

QUESTION 10

In rabbits, black fur is caused by a dominant allele B, while brown fur is caused by the recessive allele b.

Two black rabbits produce a brown offspring.

What is the probability that their **next two offspring are both black**?

A $\frac{1}{4}$

B $\frac{3}{8}$

C $\frac{1}{2}$

D $\frac{9}{16}$

E $\frac{3}{4}$

WORKED SOLUTION

A brown offspring must have genotype bb, so each black parent must have contributed a recessive allele. The parental cross is therefore

$$Bb \times Bb.$$

The probability that one offspring is black is $\frac{3}{4}$.

For two successive offspring:

$$\frac{3}{4} \times \frac{3}{4} = \boxed{\frac{9}{16}}.$$

Therefore the answer is **D**.

B-11

DNA and mutation

Answer **B**

QUESTION 11

The relevant coding-DNA triplets are shown.

A section of coding DNA is

$$\text{ATG CCT GAA TTT}.$$

A substitution changes the second triplet from CCT to CAT.

Which statements are correct?

1. Exactly one amino acid in the section is changed.
2. The reading frame is shifted.
3. The number of amino acids coded by the shown section must decrease.

coding-DNA triplet	amino acid
ATG	methionine
CCT	proline
CAT	histidine
GAA	glutamate
TTT	phenylalanine

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

The substitution changes the triplet from CCT to CAT, which changes proline to histidine.

A substitution neither inserts nor removes a base, so the reading frame is not shifted. No information given indicates that a stop signal has been introduced.

Thus only statement 1 is correct: **B**.

B-12

Gene technology

Answer E

QUESTION 12

A suitable copy of a human gene is inserted into a bacterial plasmid so that bacteria can produce the corresponding human protein.

Which statements are correct?

1. A restriction enzyme may be used to cut DNA.
2. DNA ligase may be used to join the human gene to plasmid DNA.
3. The recombinant plasmid must enter the nucleus of the bacterial cell before the gene can function.

- **A** none
- **B** 1 only
- **C** 2 only
- **D** 3 only
- **E** 1 and 2 only
- **F** 1 and 3 only
- **G** 2 and 3 only
- **H** 1, 2 and 3

WORKED SOLUTION

Restriction enzymes can cut DNA at particular sequences, and DNA ligase can join the suitable human gene copy to plasmid DNA.

Bacteria do not possess a nucleus, so the recombinant plasmid does not have to enter one.

Therefore statements 1 and 2 only are correct: **E**.

B-13

Respiration during exercise

Answer **B**

QUESTION 13

The table shows oxygen demand and oxygen supplied to working muscle at different times during exercise.

Here, oxygen demand means the amount of oxygen that would be required to meet the muscle's energy demand entirely by aerobic respiration.

Assume the oxygen supplied is all available for aerobic respiration.

At which measured times must anaerobic respiration also be contributing to the muscle's energy supply?

time / min	oxygen demand / arbitrary units	oxygen supplied / arbitrary units
0	0.8	0.8
2	1.4	1.0
4	1.8	1.3
6	2.0	1.7
8	1.6	1.6
10	1.1	1.2

A 0 only

B 2, 4 and 6 only

C 8 only

D 2, 4, 6 and 8

E 10 only

WORKED SOLUTION

When this oxygen demand exceeds the oxygen supplied, aerobic respiration alone cannot meet the muscle's stated energy demand, so anaerobic respiration must also contribute.

This occurs at 2, 4 and 6 minutes.

At 8 minutes demand equals supply, and at 10 minutes supply is greater than demand.

Therefore the answer is **B**.

B-14

Circulation

Answer **D**

QUESTION 14

Three blood vessels have the following properties.

- Vessel P carries blood **from the heart to the lungs** and has relatively low oxygen concentration.
- Vessel Q carries blood **from the lungs to the heart** and has relatively high oxygen concentration.
- Vessel R carries oxygenated blood **from the heart to a kidney**.

Which row correctly identifies the vessels?

	P	Q	R
A	pulmonary vein	pulmonary artery	renal vein
B	aorta	vena cava	pulmonary artery
C	pulmonary artery	pulmonary vein	renal vein
D	pulmonary artery	pulmonary vein	renal artery
E	vena cava	pulmonary artery	aorta

A row A

B row B

C row C

D row D

E row E

WORKED SOLUTION

The pulmonary artery carries relatively deoxygenated blood from the heart to the lungs.

The pulmonary vein carries oxygenated blood from the lungs to the heart.

The renal artery carries oxygenated blood from the heart to a kidney.

Therefore row **D** is correct.

B-15 Digestion Answer **C**

QUESTION 15

Four tubes are incubated at $37^{\circ}\mathrm{C}$.

The pH of each tube is monitored continuously.

The lipase used has an optimum near pH 8. Bile emulsifies lipid into smaller droplets but contains no digestive enzyme.

Which tube would be expected to show the **greatest initial rate of decrease in pH**?

tube	contents
A	lipid + water
B	lipid + lipase at pH 8
C	lipid + lipase + bile at pH 8
D	lipid + lipase + bile at pH 2

A A

B B

C C

D D

E All at the same rate

WORKED SOLUTION

Lipase hydrolyses lipids to products including fatty acids, so lipase activity causes the pH to fall.

Tubes B and C are at pH 8, near the stated optimum for the lipase. Tube C also contains bile, which emulsifies the lipid into smaller droplets and increases the surface area available to the enzyme. Tube D is at pH 2, far from the stated optimum, and tube A contains no lipase.

Tube C therefore has the conditions expected to give the greatest initial rate of decrease in pH. The answer is **C**.

B-16

Kidney function

Answer E

QUESTION 16

The table gives relative concentrations of substances in three fluids from a healthy person.

Which statements are correct?

1. Most plasma proteins are prevented from entering the filtrate.
2. Glucose is reabsorbed from the nephron.
3. The greater concentration of urea in urine proves that urea is produced by kidney cells.

substance	blood plasma	filtrate entering nephron	urine
protein	8.0	0	0
glucose	1.0	1.0	0
urea	0.3	0.3	2.0

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

The absence of protein from the filtrate is consistent with large plasma proteins being retained in the blood.

Glucose is present in the filtrate but absent from normal urine, supporting its reabsorption.

A greater urea concentration in urine does not show that the kidney produces urea; water removal can concentrate urea already present.

Therefore statements 1 and 2 only are correct: **E**.

B-17

Nervous system

Answer **E**

QUESTION 17

Which statements about a simple reflex arc are correct?

1. A sensory neurone can carry an impulse from a receptor towards the central nervous system.
2. A relay neurone can connect a sensory neurone to a motor neurone within the central nervous system.
3. A motor neurone normally carries impulses from an effector towards the central nervous system.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Sensory neurones carry impulses from receptors towards the CNS. Relay neurones can connect sensory and motor neurones within the CNS.

Motor neurones carry impulses from the CNS to effectors, not from effectors back towards the CNS.

Therefore statements 1 and 2 only are correct: **E**.

B-18 Disease and body defence Answer H

QUESTION 18

Which statements are correct?

1. Antibiotics do not directly destroy viruses.
2. Vaccination may lead to the formation of memory cells.
3. Antibiotic treatment can favour the survival and reproduction of resistant bacteria already present in a population.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

All three statements are correct.

Antibiotics target bacteria rather than directly destroying viruses. Vaccination can produce memory cells. Antibiotic exposure also acts as a selection pressure, allowing resistant bacteria to survive and reproduce more successfully.

Therefore the answer is **H**.

B-19 Ecosystems Answer **B**

QUESTION 19

Students record four species along a transect.

Which statements are correct?

1. Species richness is greatest at both 10 m and 20 m.
2. Total abundance is greatest at 30 m.
3. Species C occurs at every sampled position.

distance / m	species A	species B	species C	species D
0	4	0	2	0
10	3	1	2	0
20	0	4	3	1
30	0	2	0	5

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Species richness is the number of different species present. At both 10 m and 20 m, three species are present, the greatest richness recorded.

Total abundance is $6, 6, 8, 7$ at 0, 10, 20 and 30 m respectively, so statement 2 is false.

Species C is absent at 30 m, so statement 3 is false.

Thus statement 1 only is correct: **B**.

B-20 Reproductive hormones Answer **C**

QUESTION 20

Hormones P, Q and R have the following roles.

- P is released from the pituitary gland and stimulates development of an ovarian follicle.
- Q is released from the pituitary gland and its rise triggers ovulation.
- R is released from the ovary after ovulation and helps maintain the uterine lining.

Which row correctly identifies P, Q and R?

	P	Q	R
A	FSH	progesterone	LH
B	LH	FSH	progesterone
C	FSH	LH	progesterone
D	oestrogen	LH	FSH
E	FSH	LH	oestrogen

A row A

B row B

C row C

D row D

E row E

WORKED SOLUTION

FSH stimulates development of an ovarian follicle. LH triggers ovulation. Progesterone is produced after ovulation and helps maintain the uterine lining.

Therefore the correct row is **C**.

B-21

Photosynthesis and limiting factors

Answer **E**

QUESTION 21

A plant is exposed to different light intensities and carbon dioxide concentrations. All other conditions are kept constant.

Which statements are supported by the data?

1. At low light intensity, light could be the limiting factor.
2. At high light intensity and 0.04% carbon dioxide, carbon dioxide could be a limiting factor.
3. Increasing light intensity and carbon dioxide concentration indefinitely must continue increasing the rate.

light intensity	carbon dioxide concentration	rate of photosynthesis / arbitrary units
low	0.04%	5
low	0.10%	5
high	0.04%	8
high	0.10%	14

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

At low light intensity, increasing carbon dioxide does not increase the rate, so light could be limiting.

At high light intensity, increasing carbon dioxide from 0.04% to 0.10% increases the rate, so carbon dioxide could be limiting at 0.04%.

The rate cannot be assumed to increase indefinitely because another factor will eventually become limiting.

Therefore statements 1 and 2 only are correct: **E**.

B-22

Transpiration

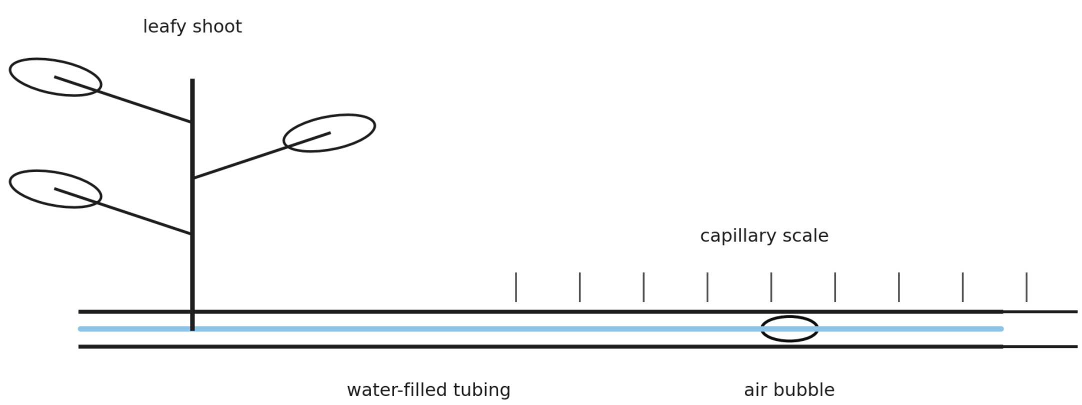
Answer **D**

QUESTION 22

A potometer is used to estimate water uptake by a leafy shoot.

The air bubble moves $18\ \mathrm{mm}$ in 6 minutes. The cross-sectional area of the capillary tube is $0.50\ \mathrm{mm}^2$.

Which row gives the rate of water uptake and the likely effect of increasing the humidity around the leaves?



	water uptake / $mm^3\ min^{-1}$	movement of bubble when humidity increases
A	0.75	faster
B	0.75	slower
C	1.50	faster
D	1.50	slower
E	3.00	slower

A row A

B row B

C row C

D row D

E row E

WORKED SOLUTION

The volume of water taken up is

$$18 \times 0.50 = 9.0\ \mathrm{mm}^3.$$

Over 6 minutes, the rate is

$$\frac{9.0}{6} = 1.50\ \mathrm{mm}^3\ \mathrm{min}^{-1}.$$

Increasing humidity reduces the water-vapour concentration gradient between the leaf and surrounding air, so transpiration decreases and the bubble moves more slowly.

Therefore row **D** is correct.

B-23

Translocation

Answer B

QUESTION 23

A plant is exposed to carbon dioxide containing radioactive carbon.

Several hours later, radioactive sucrose is detected in the roots.

Which explanation is correct?

- **A** Radioactive carbon dioxide entered the roots and was converted directly to sucrose.
- **B** Radioactive carbon dioxide entered photosynthetic tissue, was incorporated into sugars, and some sugar was transported to the roots through the phloem.
- **C** Radioactive carbon dioxide entered the leaves and was transported to the roots through the xylem before photosynthesis occurred.
- **D** Radioactive water formed in the leaves and moved through the phloem.
- **E** Radioactive carbon dioxide was absorbed directly by root hair cells by active transport.

WORKED SOLUTION

Carbon dioxide enters photosynthetic tissue and is incorporated into organic compounds during photosynthesis.

Sugars such as sucrose can then move from source tissues such as leaves to sink tissues such as roots by translocation through the phloem.

Therefore the correct answer is **B**.

B-24

Root uptake

Answer B

QUESTION 24

Root hair cells contain a higher concentration of nitrate ions than the surrounding soil solution.

Nitrate ions nevertheless continue to enter the cells.

When a chemical that inhibits respiration is added, nitrate uptake falls sharply, but water can initially continue to enter the cells.

Which row best explains these observations?

	nitrate-ion uptake	water uptake
A	diffusion	active transport
B	active transport	osmosis
C	diffusion	diffusion
D	osmosis	diffusion
E	active transport	active transport

A row A

B row B

C row C

D row D

E row E

WORKED SOLUTION

Nitrate ions are moving into cells despite already being at a higher concentration inside, so uptake is against the concentration gradient and requires active transport.

Respiration provides energy for this process, explaining why the inhibitor reduces nitrate uptake.

Water enters cells across a partially permeable membrane by osmosis.

Therefore row **B** is correct.

B-25

Surface area to volume ratio

Answer **C**

QUESTION 25

A cube of living tissue has sides of $3\ \mathrm{mm}$.

It is cut into 27 separate cubes, each with sides of $1\ \mathrm{mm}$. The total volume of tissue is unchanged.

Which row correctly describes the effect?

	total surface area after cutting compared with before	maximum diffusion distance to the centre of a cube
A	unchanged	decreases
B	doubles	unchanged
C	triples	decreases
D	increases sixfold	increases
E	increases ninefold	decreases

A row A

B row B

C row C

D row D

E row E

WORKED SOLUTION

The original surface area is

$$6(3^2) = 54\ \mathrm{mm}^2.$$

Each $1\ \mathrm{mm}$ cube has surface area $6\ \mathrm{mm}^2$, so 27 cubes have total surface area

$$27(6) = 162\ \mathrm{mm}^2.$$

Thus surface area triples:

$$\frac{162}{54} = 3.$$

The maximum distance from the surface to the centre falls from $1.5\ \mathrm{mm}$ to $0.5\ \mathrm{mm}$.

Therefore row **C** is correct.

B-26 Meiosis and sex determination Answer **F**

QUESTION 26

A cell in the testes of a healthy human male undergoes meiosis to produce sperm.

Which statements are correct?

1. Each normal sperm produced contains 23 chromosomes.

2. Every sperm produced contains a Y chromosome.

3. Fertilisation of an egg by a Y-bearing sperm can produce an XY zygote.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Meiosis produces haploid gametes containing 23 chromosomes.

Sperm contain either an X chromosome or a Y chromosome, so statement 2 is false.

A Y-bearing sperm fertilising an X-bearing egg can produce an XY zygote.

Therefore statements 1 and 3 only are correct: **F**.

B-27 Blood glucose control Answer **E**

QUESTION 27

After a carbohydrate-rich meal, the blood glucose concentration of a healthy person rises.

Which statements are correct?

1. Insulin secretion should increase.
2. Increased uptake of glucose by target tissues can contribute to the return of blood glucose concentration towards its normal level.
3. Glucagon secretion should increase because glucagon promotes uptake of glucose from the blood into the liver.

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

A rise in blood glucose concentration stimulates increased insulin secretion.

Insulin promotes processes that remove glucose from the blood, helping return the concentration towards its normal range by negative feedback.

Glucagon acts in the opposite direction when blood glucose is low; it promotes processes that increase blood glucose concentration.

Therefore statements 1 and 2 only are correct: **E**.

Compilation record: Level 1 uses Engineering Mock Test 3 for Mathematics/Physics and the corresponding Standard Biology/Chemistry source sets. This solution book is ready to support the later construction of the matching full test.